



Doubt Clearing Session

Complete Course on Computer Architecture for GATE



Sanchit Jain

Legend

I am a computer science faculty, having a teaching experience of more than 8 years. I have a very sound experience in teaching students.

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3M Watch mins (last 30 days)

21K Followers

2K Dedications



Educator highlights

-  **Works at Knowledge Gate**
-  **Studied at Delhi Technological University**
-  **Have qualified gate Have experience of more than 8 years Have a YouTube channel with 5 lakh follower**
-  **Lives in Ghaziabad, Uttar Pradesh, India**
-  **Unacademy Educator since 19th July, 2019**
-  **844,388 live minutes taught in last 30 days**
-  **Knows Punjabi, Hinglish, Hindi and English**

Months Wise Calendar for GATE-2022 (Unacademy Plus) (Sanchit Jain)

Month	Morning Class (7:00am – 9:00am)	Evening Class (7:00pm – 9:00pm)
Feb-2021	DS & Programming (15-Feb to 23-Mar) (56 Hours) Algo (24-Mar to 26-April) (38 Hours) CN(29-April to 15-June) (54 Hours) TOC (26-May to 05-July) (58 Hours) Compiler (7-July to 5-Aug) (34 Hours)	DM (22-Feb to 27-Mar) (50 Hours) DBMS (29-Mar to 30-April) (50 Hours) DE (7-May to 1-June) (40 Hours) COA (23-June to 20-July) (38 Hours) OS (21-July to 24-Aug) (58 Hours)
Mar-2021		
Apr-2021		
May-2021		
Jun-2021		
Jul-2021		
Aug-2021	DM (11-Aug to 20-Sept) (50 Hours) DBMS (22-Sept to 29-Oct) (50 Hours) DE (3-Nov to 23-Nov) (40 Hours) COA (24-Nov to 13-Dec) (38 Hours) OS (15-Dec to 11-Jan) (58 Hours) TOC (12-Jan to 12-Feb) (58 Hours)	DS & Programming (25-Aug to 5-Oct) (56 Hours) Algo (6-Oct to 1-Nov) (38 Hours) CN(10-Nov to 5 Jan) (54 Hours) Compiler (26-Jan to 15-Feb) (34 Hours)
Sep-2021		
Oct-2021		
Nov-2021		
Dec-2021		
Jan-2022		
Feb-2022		

Date	Day	Subject-1	Topic	Time
24-11-2021	Wednesday	COA	memory hierarchy	7:00 am to 9:00 am
25-11-2021	Thursday	COA	Cache Mapping Part-1	7:00 am to 9:00 am
26-11-2021	Friday	COA	Cache Mapping Part-2	7:00 am to 9:00 am
27-11-2021	Saturday	COA	Doubt Clearing Session	7:00 am to 9:00 am
28-11-2021	Sunday	COA	Cache Management Part-1	7:00 am to 9:00 am
29-11-2021	Monday	COA	Cache Management Part-2	7:00 am to 9:00 am
30-11-2021	Tuesday			
01-12-2021	Wednesday	COA	Input-Output Organization Part-1	7:00 am to 9:00 am
02-12-2021	Thursday	COA	Doubt Clearing Session	7:00 am to 9:00 am
03-12-2021	Friday	COA	Input-Output Organization Part-2	7:00 am to 9:00 am
04-12-2021	Saturday	COA	Input-Output Organization Part-3	7:00 am to 9:00 am
05-12-2021	Sunday	COA	Input-Output Organization Part-4	7:00 am to 9:00 am
06-12-2021	Monday	COA	Doubt Clearing Session	7:00 am to 9:00 am
07-12-2021	Tuesday	COA	Pipelining Part-1	7:00 am to 9:00 am
08-12-2021	Wednesday	COA	Pipelining Part-2	7:00 am to 9:00 am
09-12-2021	Thursday	COA	Instruction Formate and Addressing mode Part-1	7:00 am to 9:00 am
10-12-2021	Friday	COA	Doubt Clearing Session	7:00 am to 9:00 am
11-12-2021	Saturday	COA	Instruction Formate and Addressing mode Part-2	7:00 am to 9:00 am
12-12-2021	Sunday	COA	Control Unit Part-1	7:00 am to 9:00 am
13-12-2021	Monday	COA	Control Unit Part-2	7:00 am to 9:00 am

Date	Day	Subject-2	Topic	Time	Duration
10-11-2021	Wednesday	CN	Basics of Computer Networks Part-1	6:00 pm to 8:00 pm	2
11-11-2021	Thursday	CN	Basics of Computer Networks Part-2	6:00 pm to 8:00 pm	2
15-11-2021	Monday	CN	Basics of Computer Networks Part-3	6:00 pm to 8:00 pm	2
16-11-2021	Tuesday	CN	Doubt Clearing Session	6:00 pm to 8:00 pm	2
17-11-2021	Wednesday	CN	Access Control Part-1	6:00 pm to 8:00 pm	2
18-11-2021	Thursday	CN	Access Control Part-2	6:00 pm to 8:00 pm	2
22-11-2021	Monday	CN	Flow Control Part-1	6:00 pm to 8:00 pm	2
23-11-2021	Tuesday	CN	Doubt Clearing Session	6:00 pm to 8:00 pm	2
24-11-2021	Wednesday	CN	Error Control Part-1	6:00 pm to 8:00 pm	2
25-11-2021	Thursday	CN	Error Control Part-2	6:00 pm to 8:00 pm	2
29-11-2021	Monday	CN	Framing and Ethernet	6:00 pm to 8:00 pm	2
30-11-2021	Tuesday	CN	Doubt Clearing Session	6:00 pm to 8:00 pm	2
01-12-2021	Wednesday	CN	Network Layer protocol Part-1	6:00 pm to 8:00 pm	2
02-12-2021	Thursday	CN	Network Layer protocol Part-2	6:00 pm to 8:00 pm	2
06-12-2021	Monday	CN	Network Layer protocol Part-3	6:00 pm to 8:00 pm	2
07-12-2021	Tuesday	CN	Doubt Clearing Session	6:00 pm to 8:00 pm	2
13-12-2021	Monday	CN	IP addressing and routing and Routing Part-1	6:00 pm to 8:00 pm	2
14-12-2021	Tuesday	CN	IP addressing and routing and Routing Part-2	6:00 pm to 8:00 pm	2
20-12-2021	Monday	CN	Transport Layer Part-1	6:00 pm to 8:00 pm	2
21-12-2021	Tuesday	CN	Doubt Clearing Session	6:00 pm to 8:00 pm	2
27-12-2021	Monday	CN	Transport Layer Part-2	6:00 pm to 8:00 pm	2
28-12-2021	Tuesday	CN	Transport Layer Part-3	6:00 pm to 8:00 pm	2
03-01-2022	Monday	CN	Application Layer	6:00 pm to 8:00 pm	2
04-01-2022	Tuesday	CN	Doubt Clearing Session	6:00 pm to 8:00 pm	2
05-01-2022	Wednesday	CN	crypto	6:00 pm to 8:00 pm	2

GATE & ESE

Plus

All courses

All languages

All topics



HINGLISH ELECTRICAL ENGINEERING

Complete Course on Computer Architecture for GATE

Lesson 7 • Dec 1, 2021 7:00 AM

Sanchit Jain



6:00 PM

HINGLISH CS & IT

Complete Course on Computer Networks for GATE

Lesson 11 • Today, 6:00 PM

Sanchit Jain



HINDI CS & IT

Complete Course on Digital Electronics for GATE

Ended on Nov 22, 2021 • 18 lessons

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HINGLISH PRACTICE & STRATEGY

FAQ Handbook for GATE

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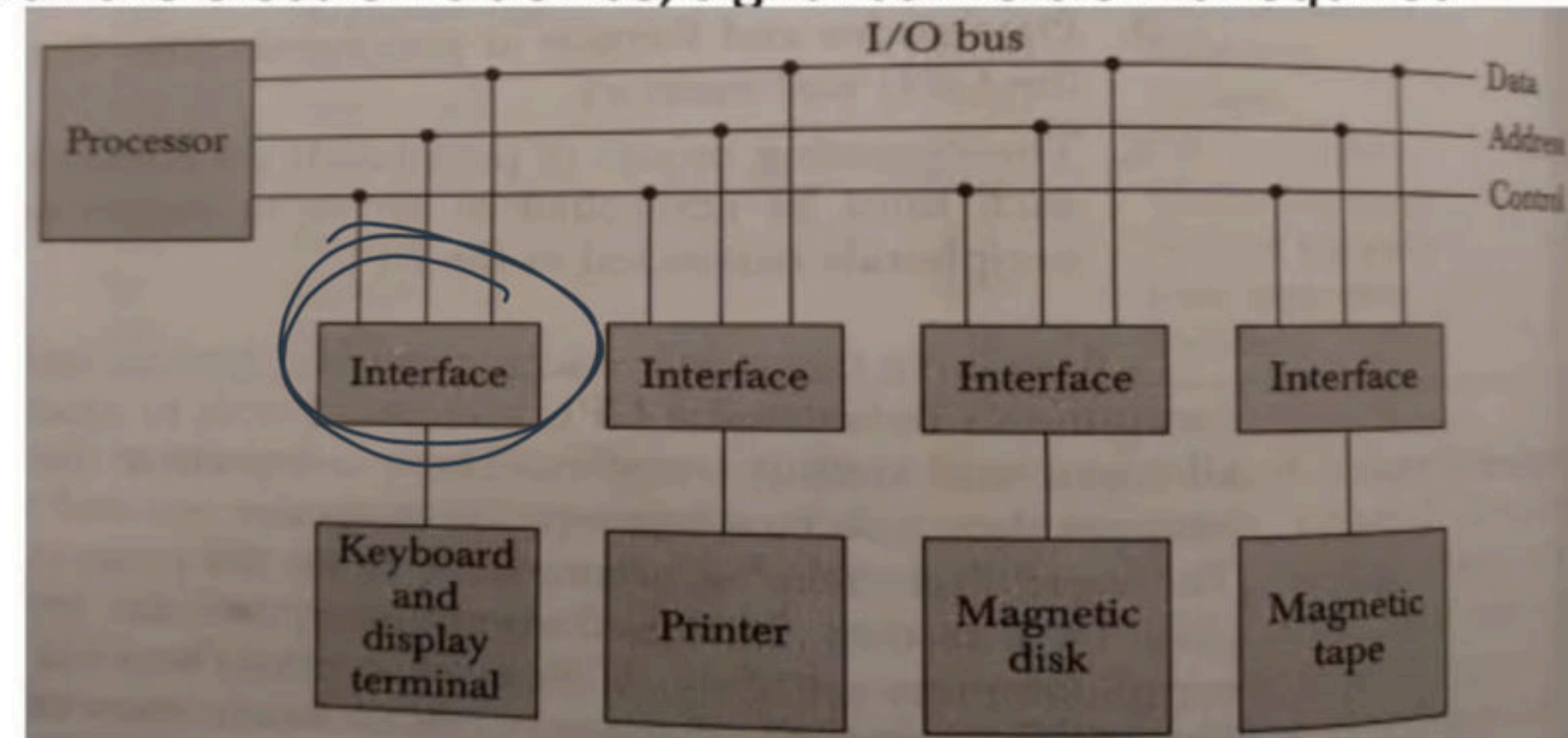
Input / Output management

- In general, when we say a computer, we understand a CPU + Memory (cache, main memory)
- But a computer does not serve any purpose if it cannot receive data from the outside world or cannot transmit the data to outside world.
- I/o or peripheral devices are those independent devices which serve this purpose.
- So, while designing i/o for a computer we must know the number of i/o device and the capacity of each device.

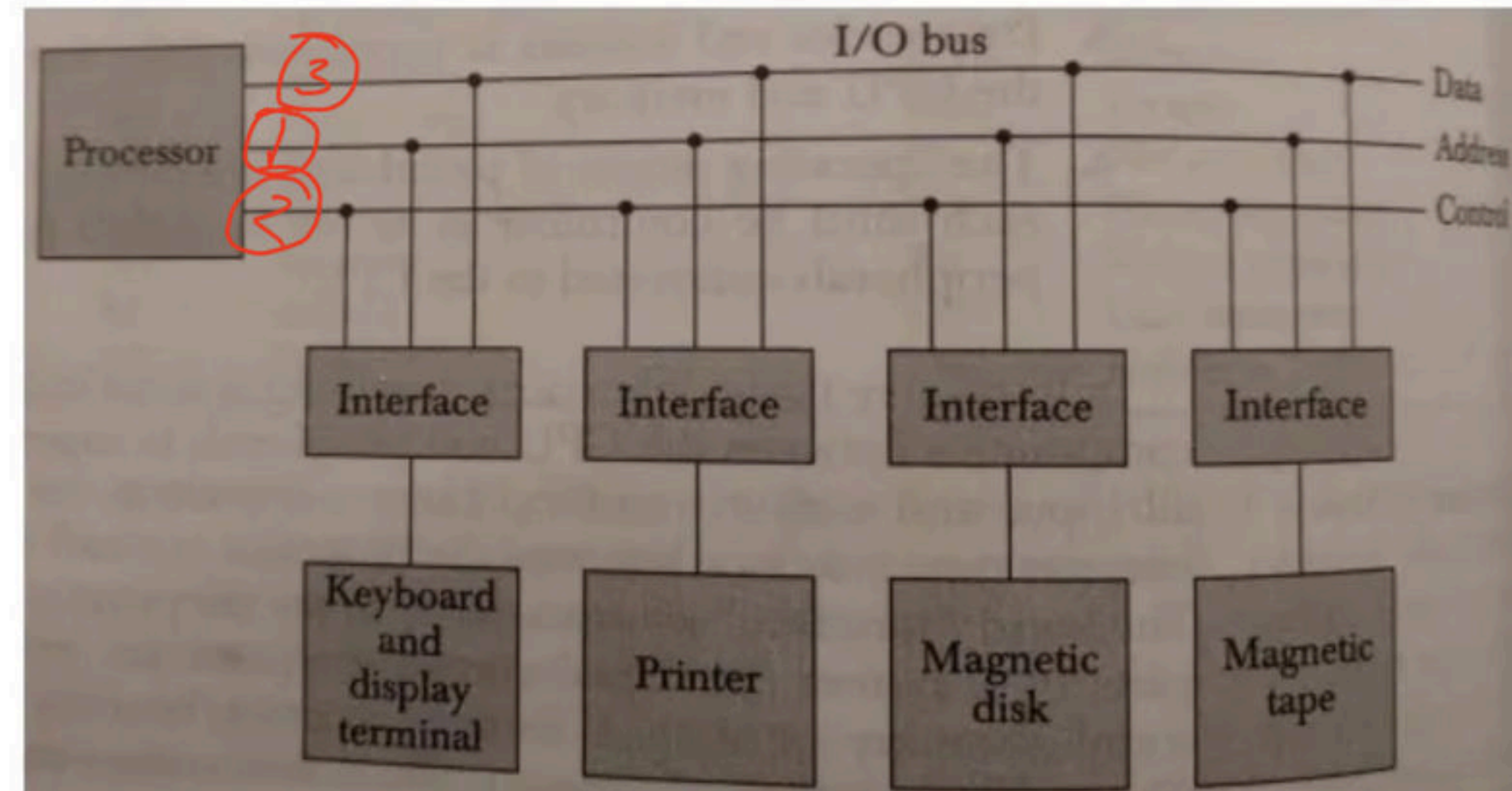
Interface

we cannot directly connect a i/o device to computer because of the following reasons

- **Speed:** - The speed of CPU and i/o device will usually different.
- **Format:** - The data code and format of CPU and peripherals may be different. E.g. ASCII, Unicode etc.
- **Physical orientation:** - Different device have organizations like optical, magnetic, electrochemical and different controlling functions.
- **Signal conversion:** - peripherals are electromagnetic and electrochemical device and their manner of operation is different from the operations of CPU and memory, which are electronic device, signal conversion is required



- **Address bus:** - Is used to identify the correct i/o devices among the number of i/o device , so CPU put an address of a specific i/o device on the address line, all devices keep monitoring this address bus and decode it, and if it is a match then it activates control and data lines.
- **Control bus:** - After selecting a specific i/o device CPU sends a functional code on the control line. The selected device(interface) reads that functional code and execute it. E.g. i/o command, control command, status command etc.
- **Data bus:** - In the final step depending on the operation either CPU will put data on the data line and device will store it or device will put data on the data line and CPU will store it.



Q Consider a main memory system that consists of 8 memory modules attached to the system bus, which is one word wide. When a write request is made, the bus is occupied for 100 nanoseconds (ns) by the data, address, and control signals. During the same 100 ns, and for 500 ns thereafter, the addressed memory module executes one cycle accepting and storing the data. The (internal) operation of different memory modules may overlap in time, but only one request can be on the bus at any time. The maximum number of stores (of one word each) that can be initiated in 1 millisecond is _____ **(Gate-2014) (2 Marks)**

(A) 1000

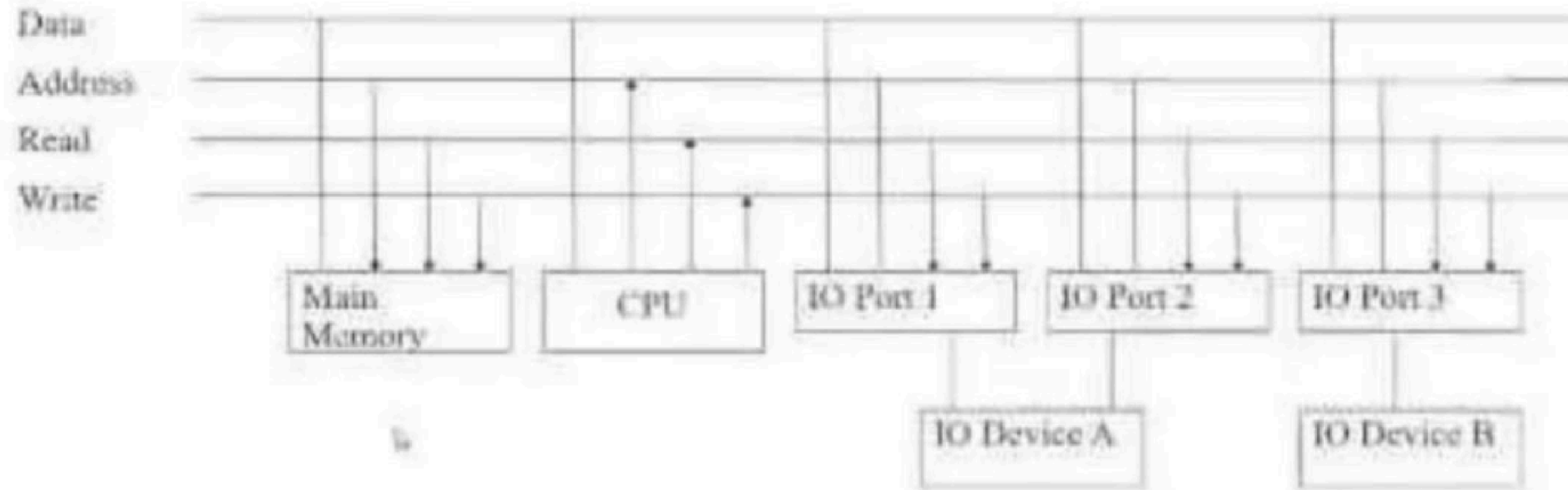
(B) 10000

(C) 100000

(D) 100

How a computer (CPU) deals with Memory and I/O devices

Memory Mapped i/o



- Here there is no separate i/o instructions. The CPU can manipulate i/o data residing in the interface registers with the same instructions that are used to manipulate memory words.
- Here computer can use memory type instructions for i/o data.
- E.g. 8085
- **Advantage:** - In typical computer there are more memory reference instruction than i/o instruction, but in memory mapped i/o all instructions that refer to memory are also available for i/o.
- **Disadvantage:** - Total address get divided, some range is occupied by i/o while some memory.

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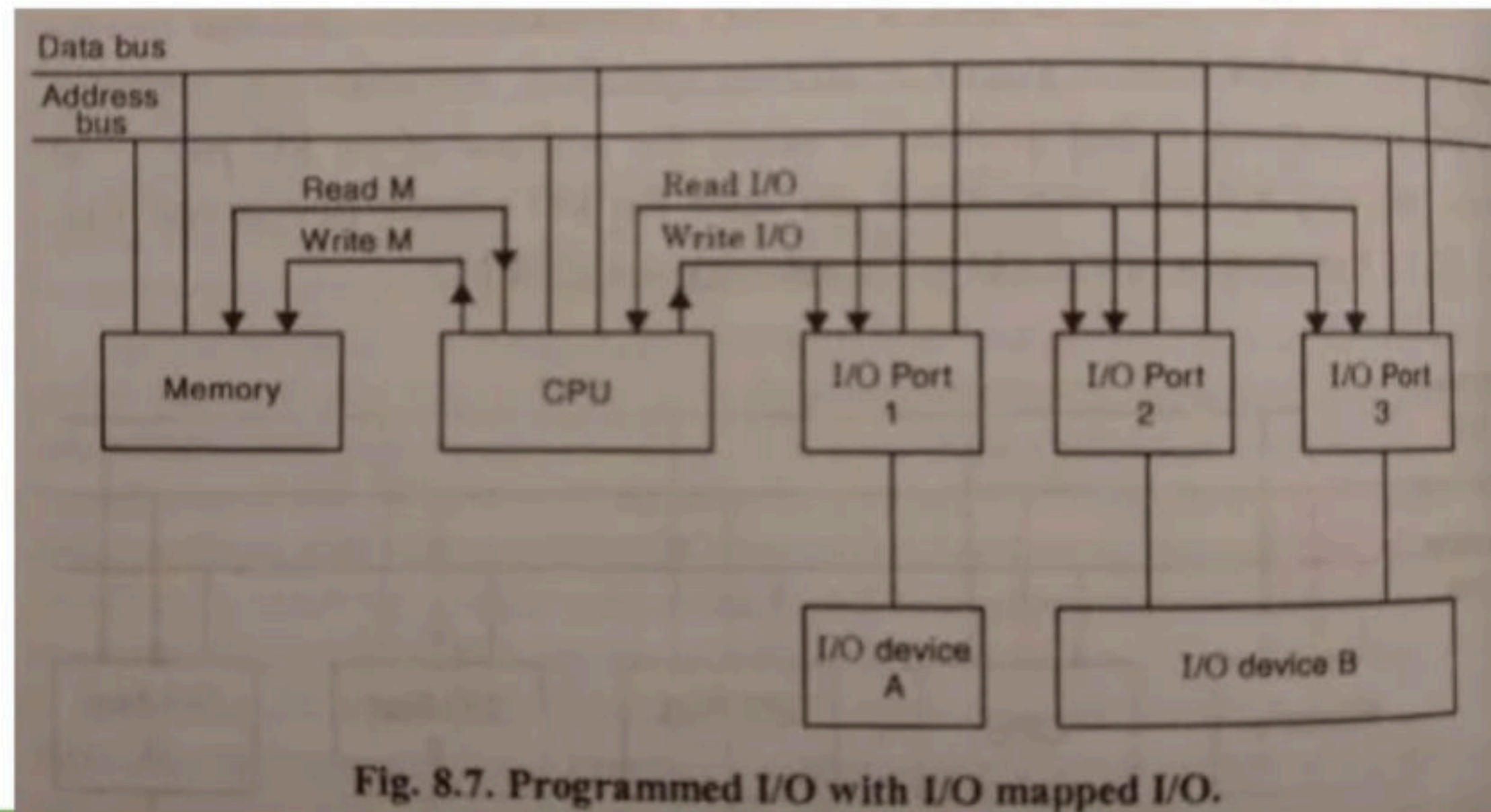
- The **Commodore 64**, also known as the **C64** or the **CBM 64**, is an 8-bit home computer introduced in January 1982 by Commodore International (first shown at the Consumer Electronics Show, in Las Vegas, January 7–10, 1982).
- It has been listed in the Guinness World Records as the highest-selling single computer model of all time, with independent estimates placing the number sold between 10 and 17 million units.



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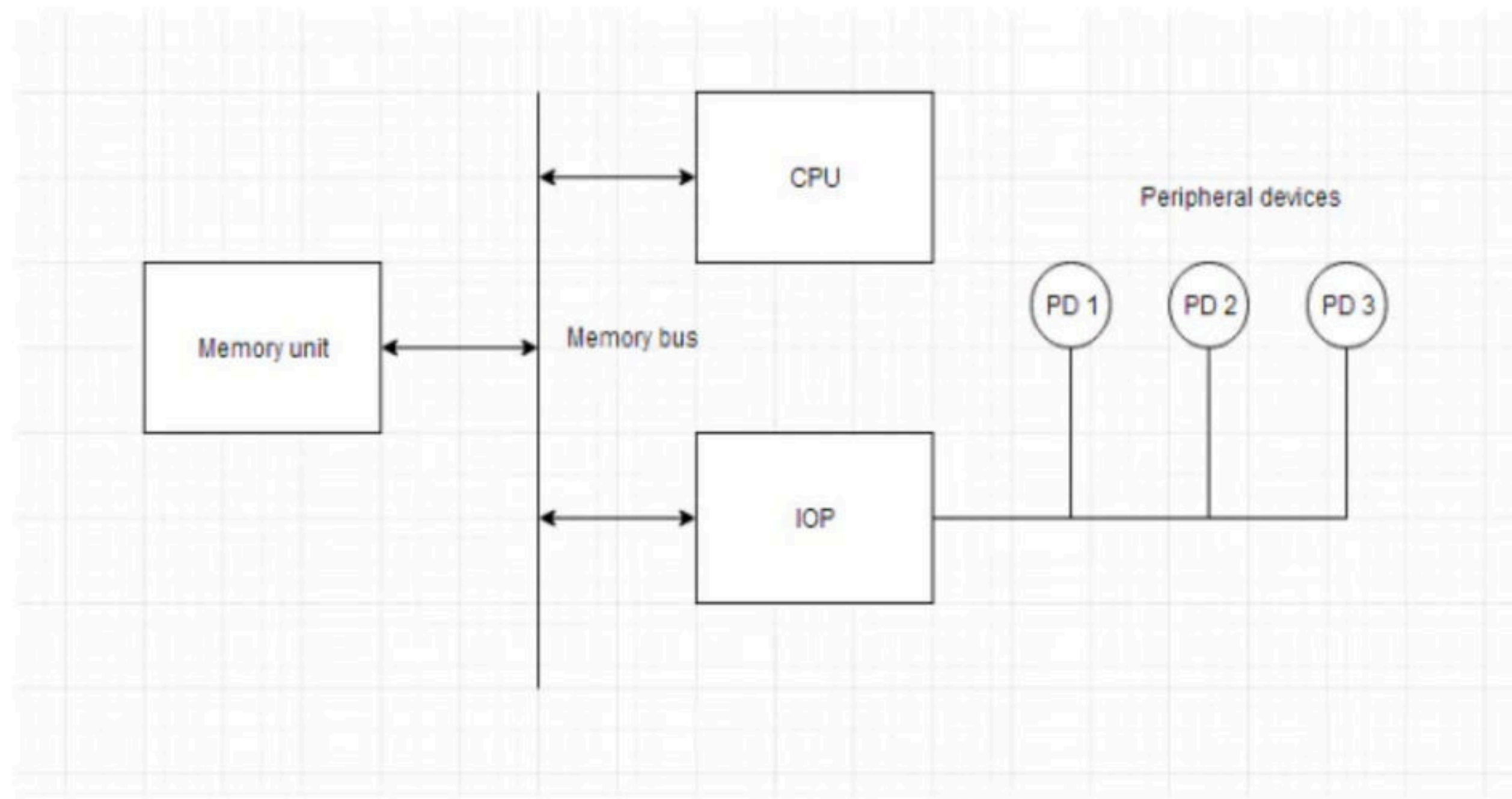
Isolated i/o

- Here common bus to transfer data between memory or i/o and CPU. The distinction between a memory and i/o transfer is made through separate read and write line.
- i/o read and i/o write control lines are enabled during an i/o transfer.
- Memory read and memory write control lines are enabled during a memory transfer.
- **Advantage:** - Here memory is used efficiently as the same address can be used two times. E.g 8086
- **Disadvantage:** - Need different control lines one for memory and other for i/o devices.



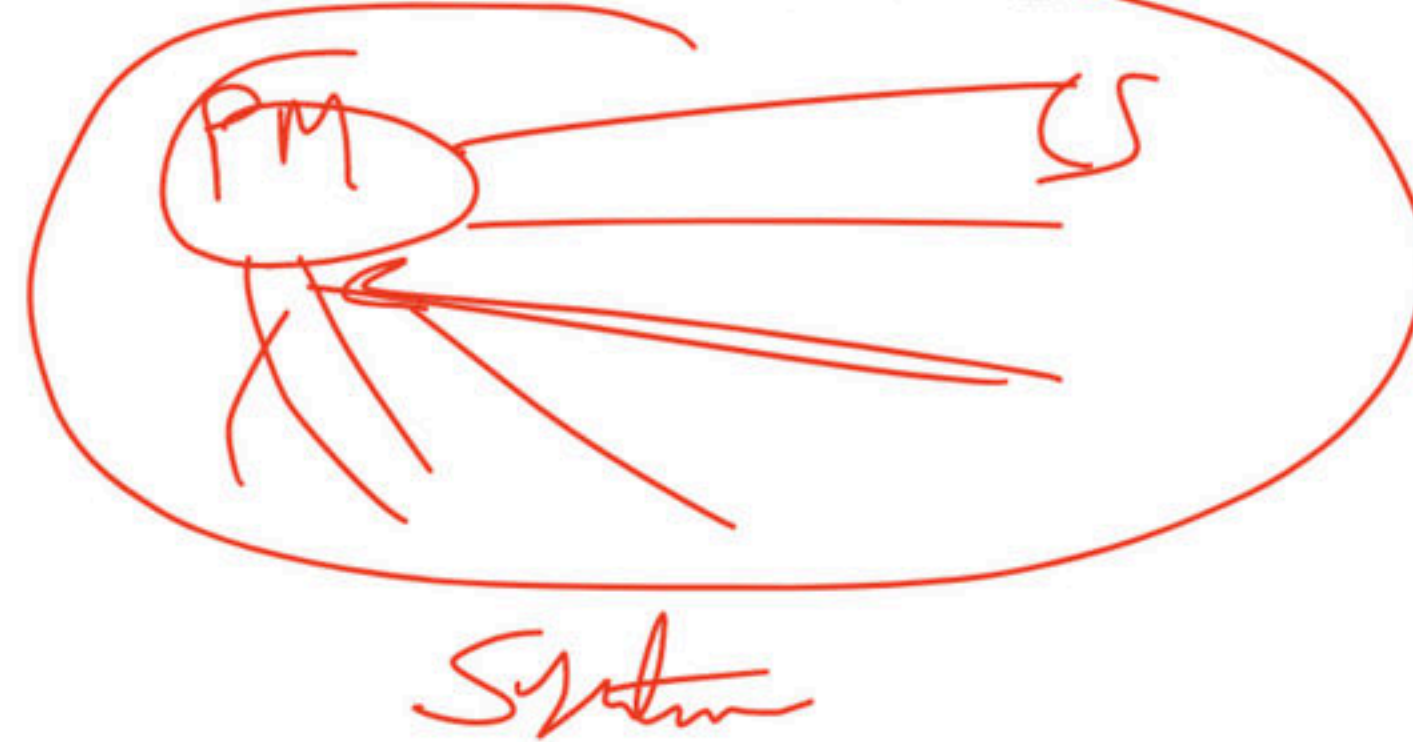
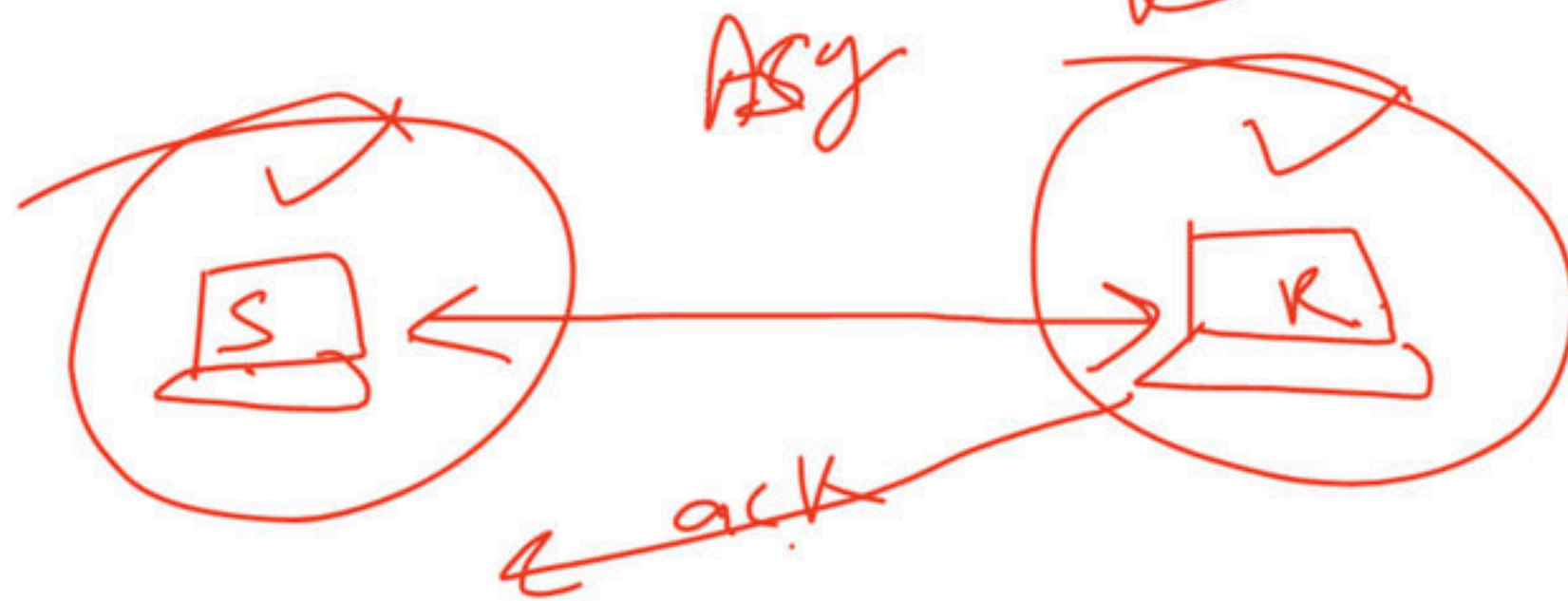
I/O Processor

- Computer has independent sets of data, address and control buses, one for accessing memory and the other for i/o. this is done in computers that provide a separate i/o processor other than CPU.
- Memory communicate with both the CPU and iop through a memory bus.
- Iop communicates also with the input and output devices through a separate i/o bus with its own address, data and control lines. The purpose of iop is to provide an independent pathway for the transfer of information between external devices and internal memory.



Synchronous Vs Asynchronous data transfer

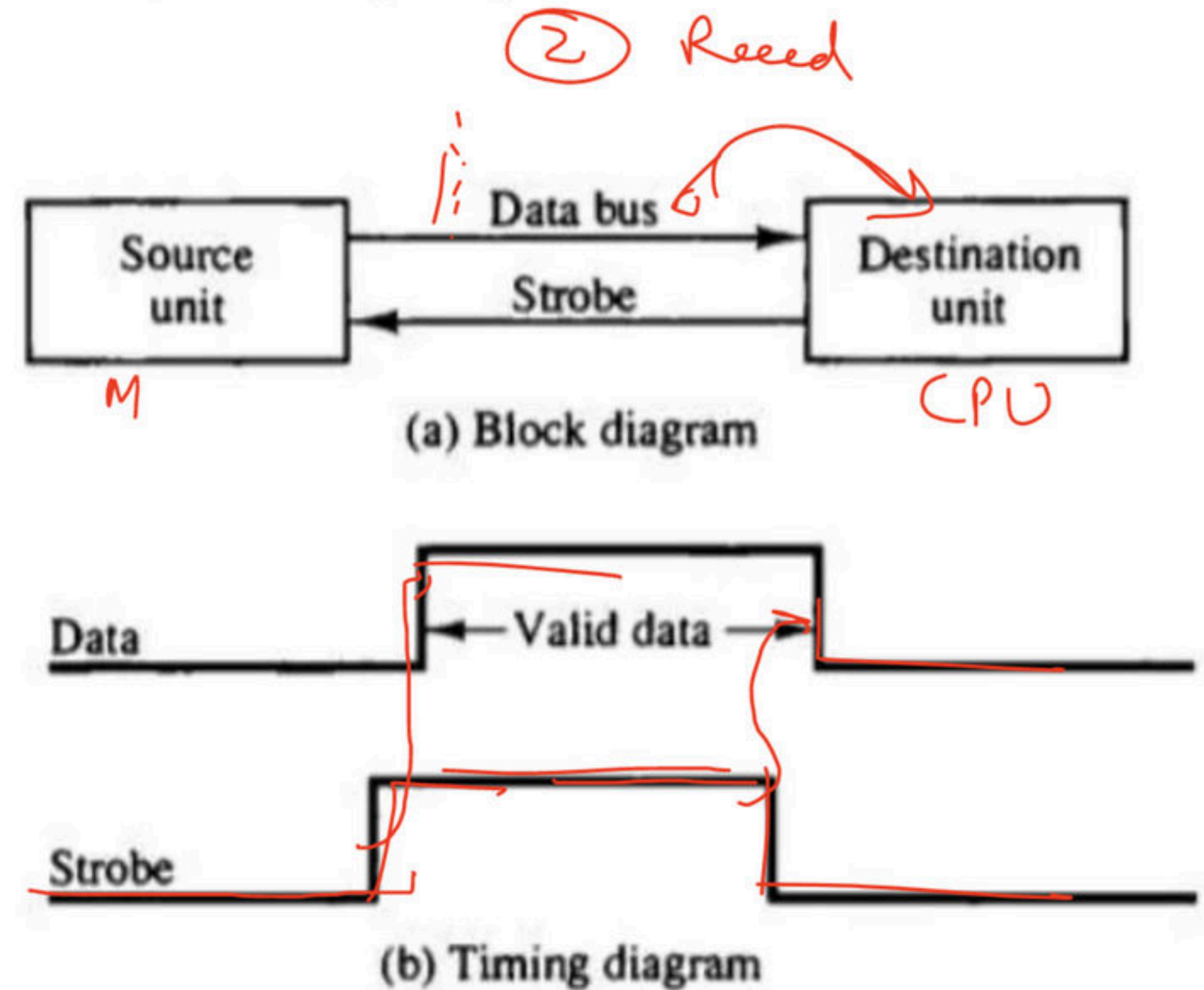
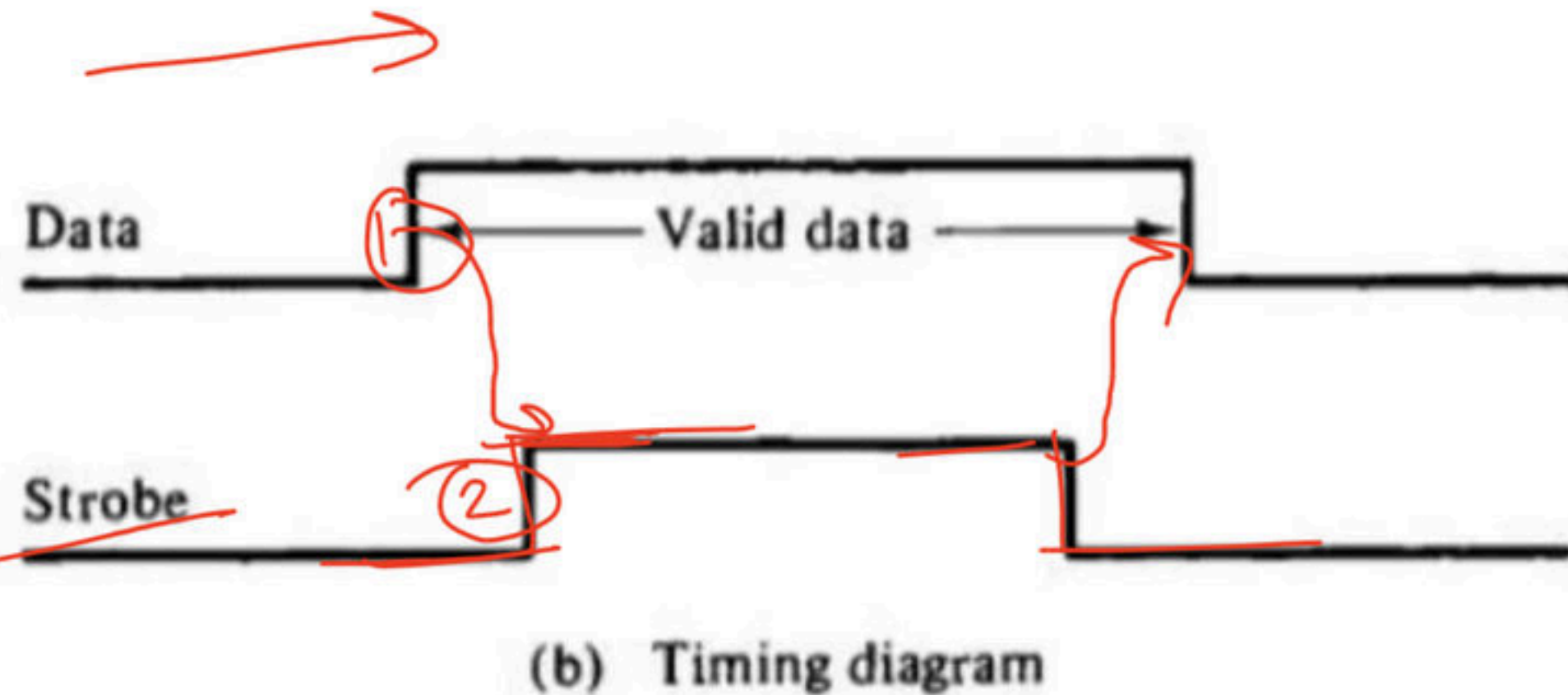
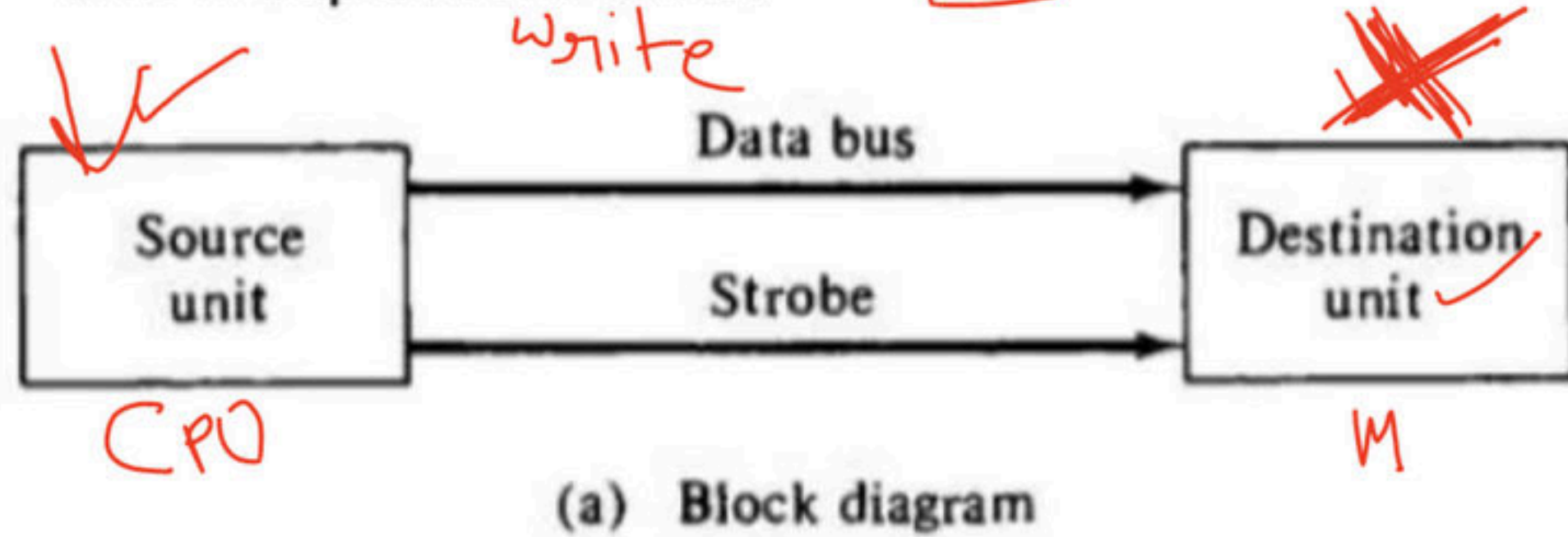
- Synchronization is achieved by a device called master generator, which generate a periodic train of clock pulse.
- The internal operations in a digital system are synchronized by means of clock pulses supplied by a common pulse generator.
- When communication happens between devices which are under a same control unit or same clock then it is called synchronous communication. e.g. communication between CPU and its registers.



Asynchronous communication

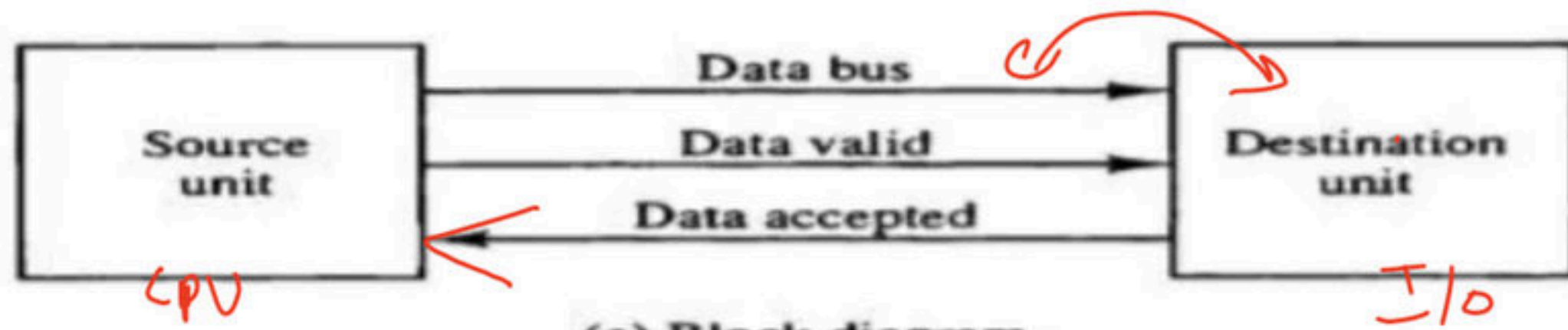
- When the timing units of two devices are independent that is they are under different control then it is called asynchronous communication.
- Asynchronous data transfer between two independent units required that control signals be transmitted between the communicating units to indicate the time at which data is being transmitted.
- One way to achieving this by means of strobe pulse supplied by one of the units to indicate to the other unit which when the transfer has to occur.

- the unit receiving the data item response with another control signal to acknowledge the receipt of the data. this type of agreement between two independent units is referred to as handshaking.
- The strobe pulse method and the handshaking method of asynchronous data transfer are not restricted to input output transfer. in fact they are used extensively on numerous occasions requiring the transfer of data between two independent units.

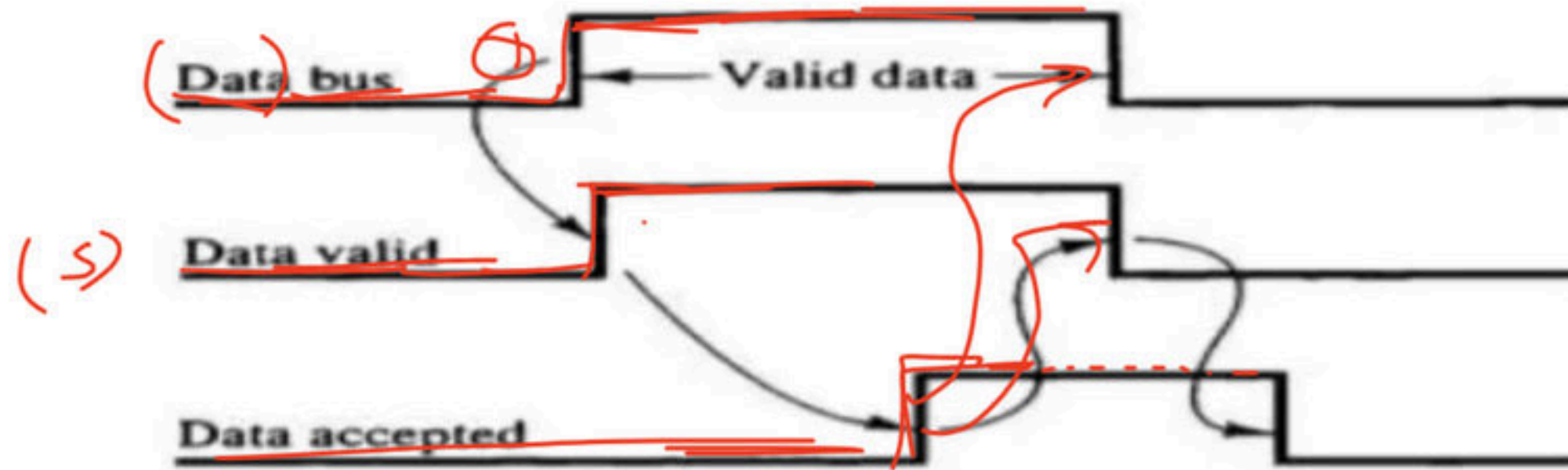


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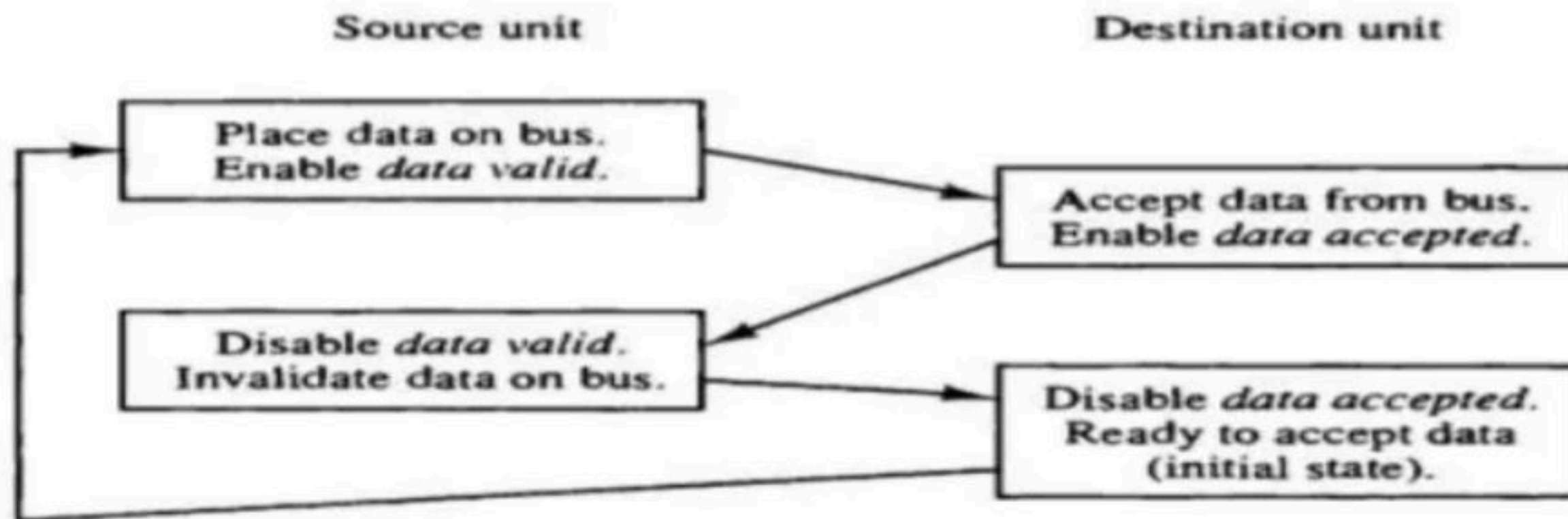
Source Initiated Transfer



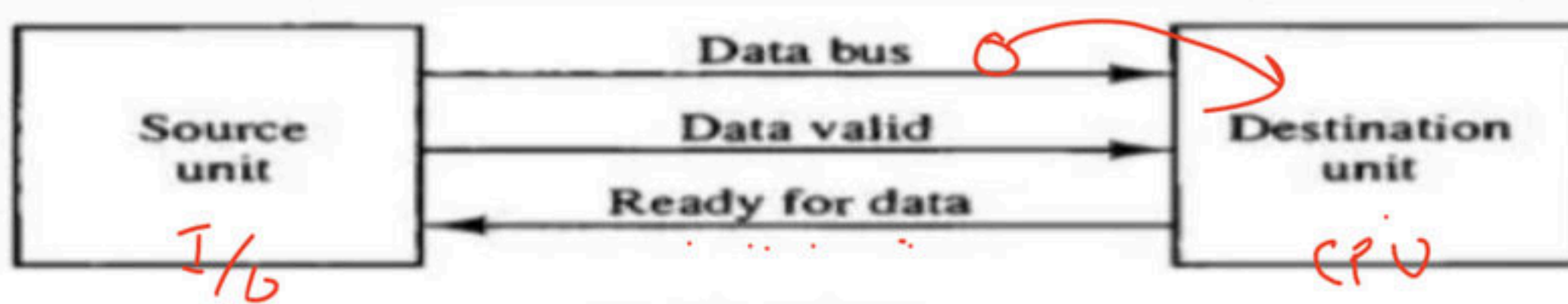
(a) Block diagram



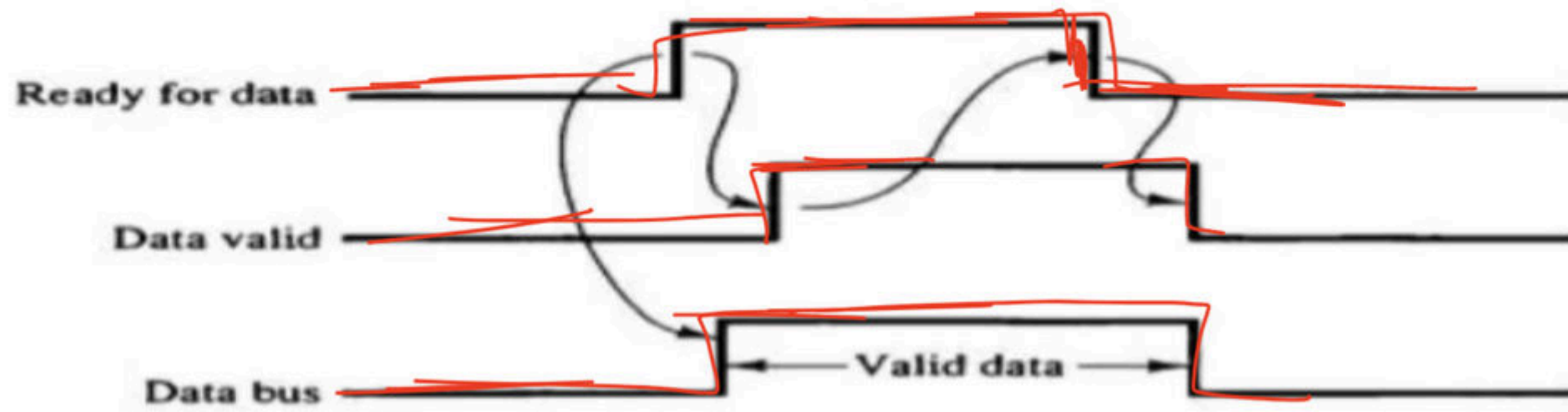
(b) Timing diagram



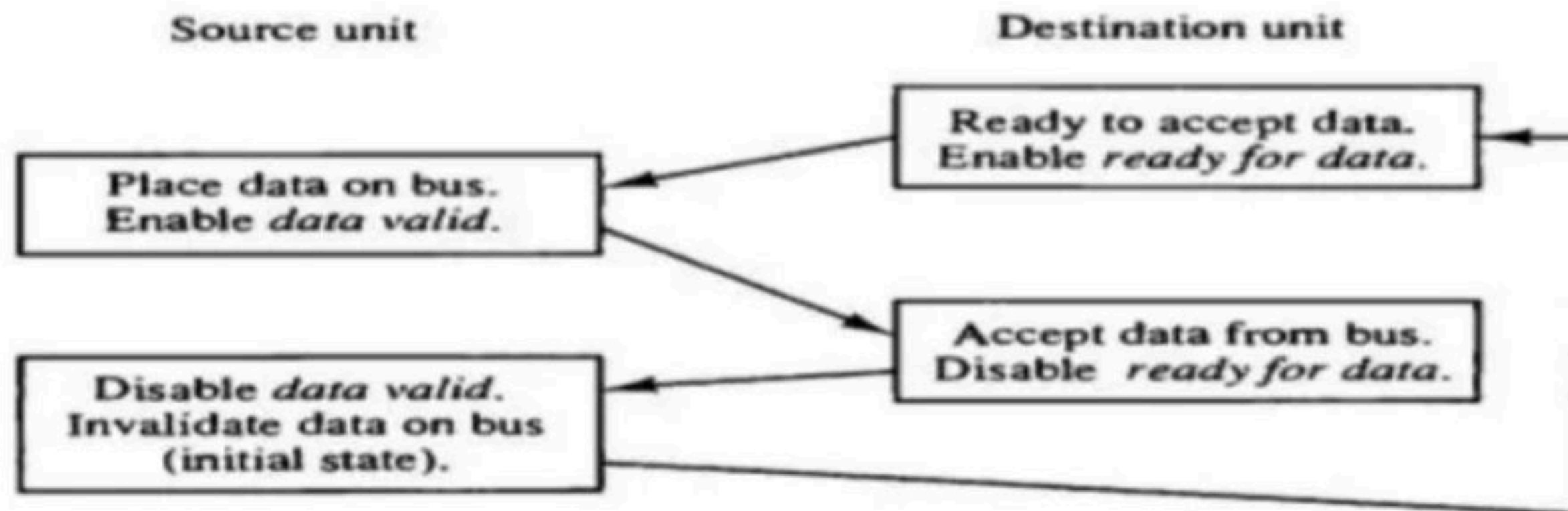
Destination initiated transfer



(a) Block diagram



(b) Timing diagram



Modes of Data Transfer

Here we deal with the problem that how data communication will take place CPU and i/o device. There are popularly three methods of data transfer: -

- ✓ • **Programmed i/o** ✓
- ✓ • **Interrupt initiated i/o**
- ✓ • **Direct Memory Transfer**

PING ms

1

DOWNLOAD Mbps

340.15

UPLOAD Mbps

333.65

GO



Connections

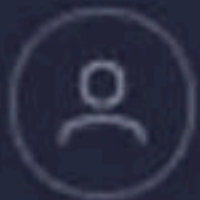
Multi



Excitel

Noida

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Excitel

205.254.171.244

HOW DOES YOUR DOWNLOAD SPEED COMPARE
WITH YOUR EXPECTATIONS?

1

2

3

4

5

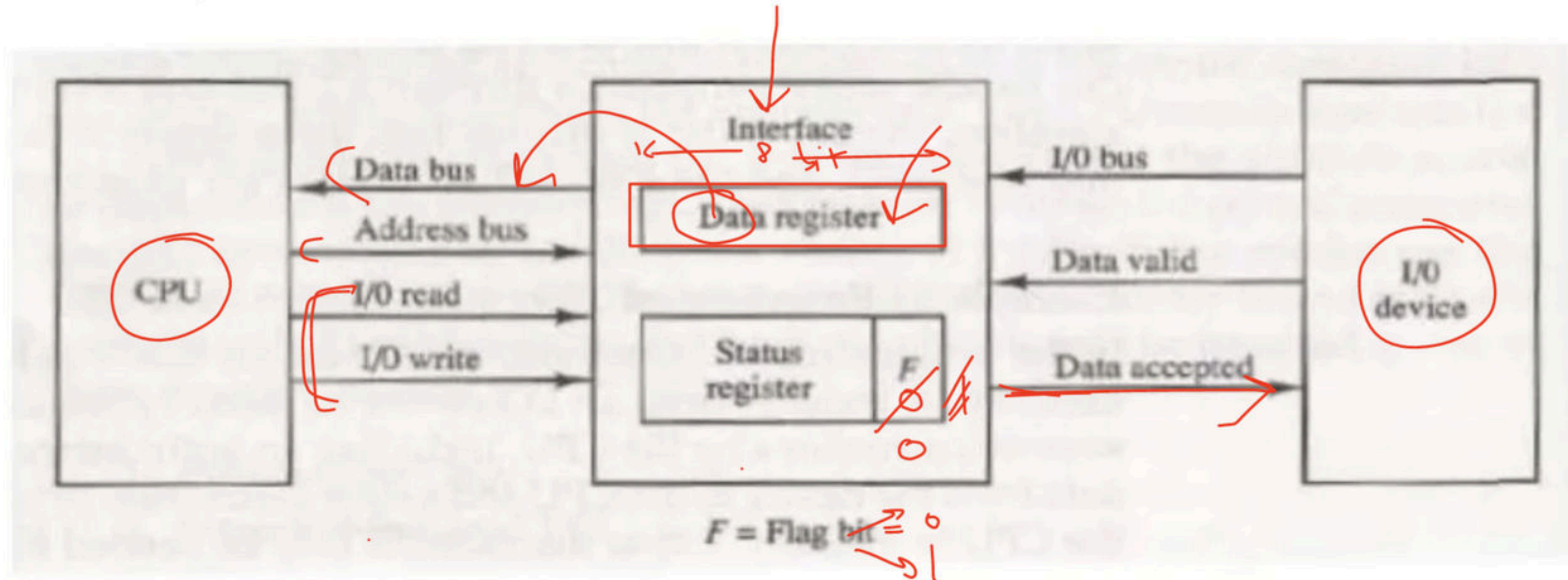
Much worse

As expected

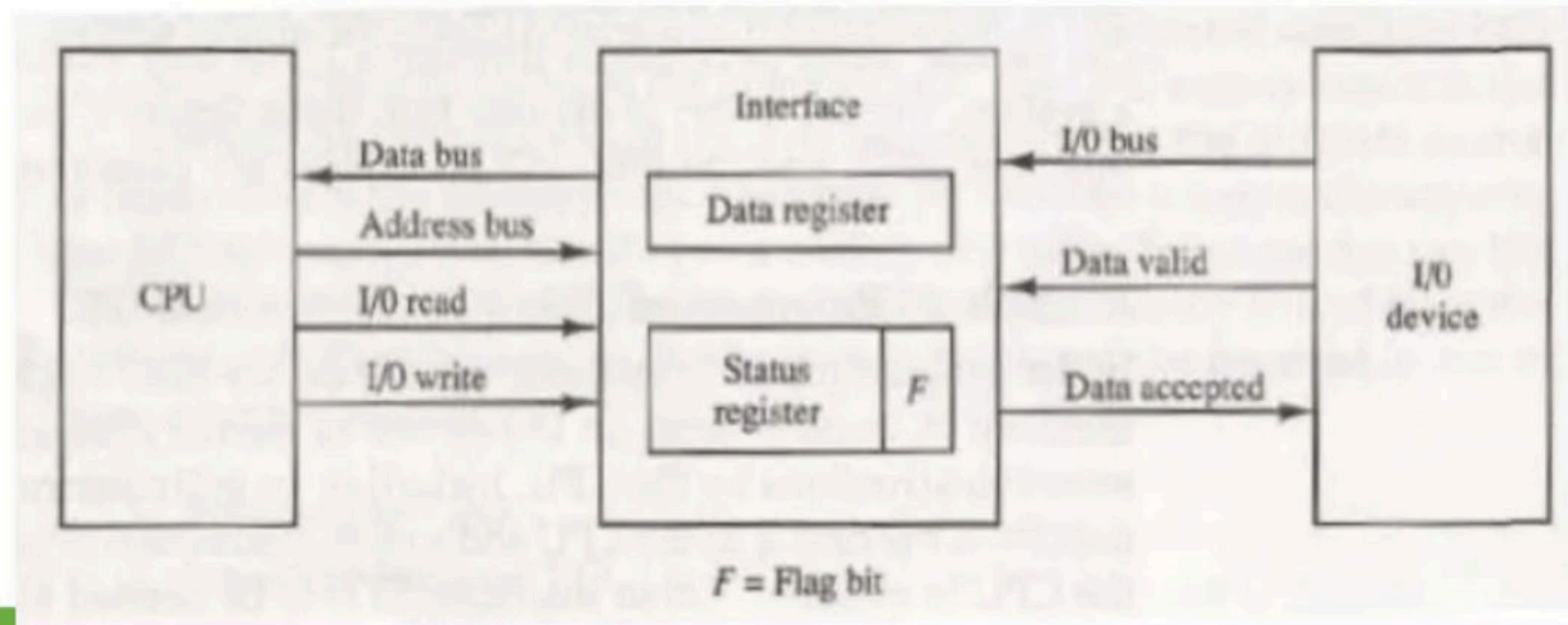
Much better

Programmed I/O

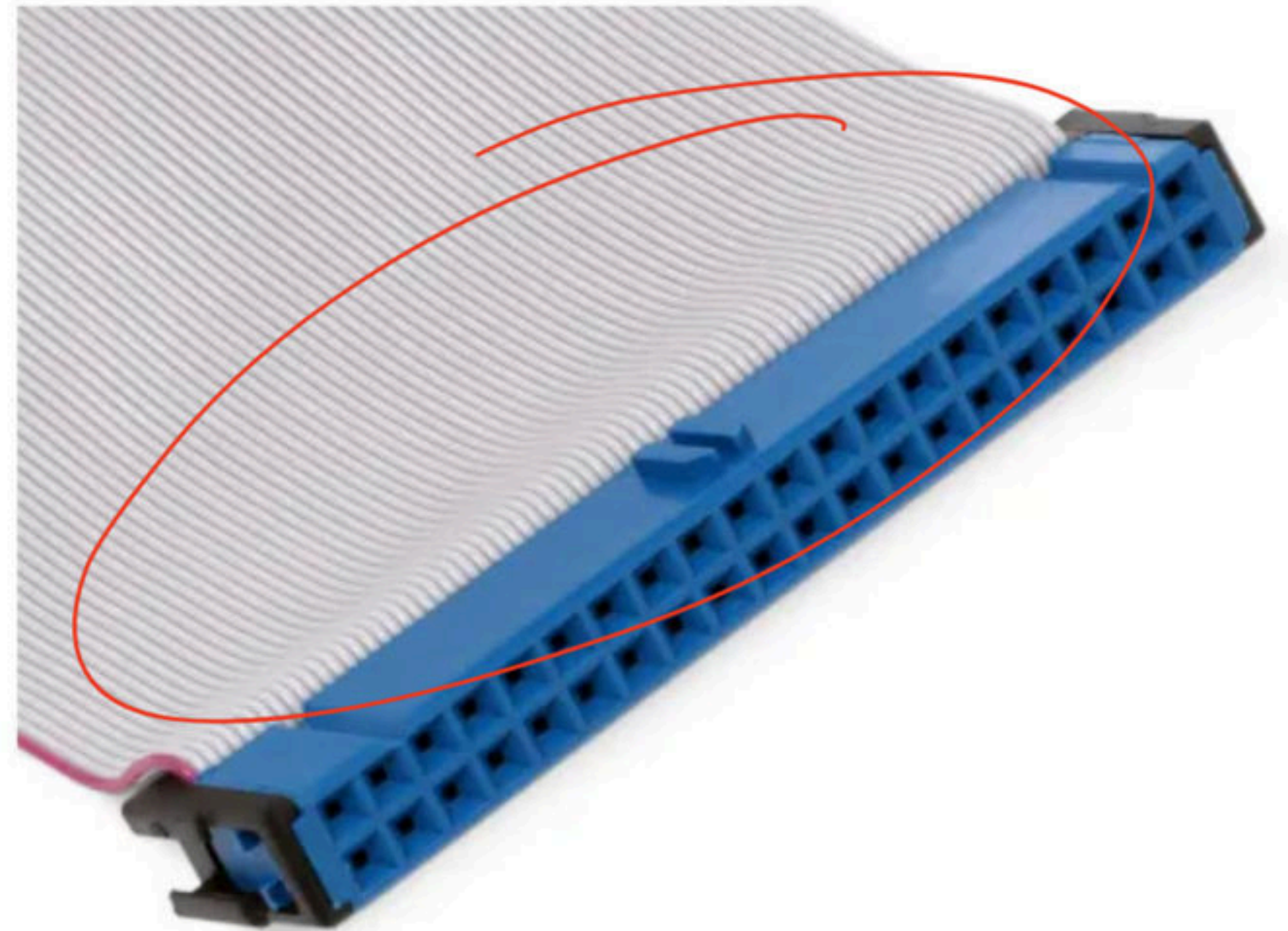
- In this i/o device cannot access the memory directly. To complete data transfer number of instructions will be executed out of which input instruction are those which transfer data from device to CPU and store instructions from CPU to memory.



- **Step 1:** - i/o device will put data on the data bus and will enable valid data signal
- **Step 2:** - Interface is continuously sensing for data valid signal and when it receives the signal it will copy data from the data bus into its data register and set its flag bit to 1 and enable data accepted line.
- **Step 3:** - CPU is continuously monitoring the status register in the programmed mode and as soon as it sees flag bit as 1, it immediately copies data from data register on the data bus and clear flag bit to zero.
- **Step 4:** - Now interface will disable data accepted line to tell i/o device, I am ready for new transitions.
- **Conclusion:** - CPU works in programmed mode or in busy wait mode so no of clock cycles are wasted. It will be difficult to handle multiple i/o device at the same time. It is not appropriate with the high-speed i/o devices.

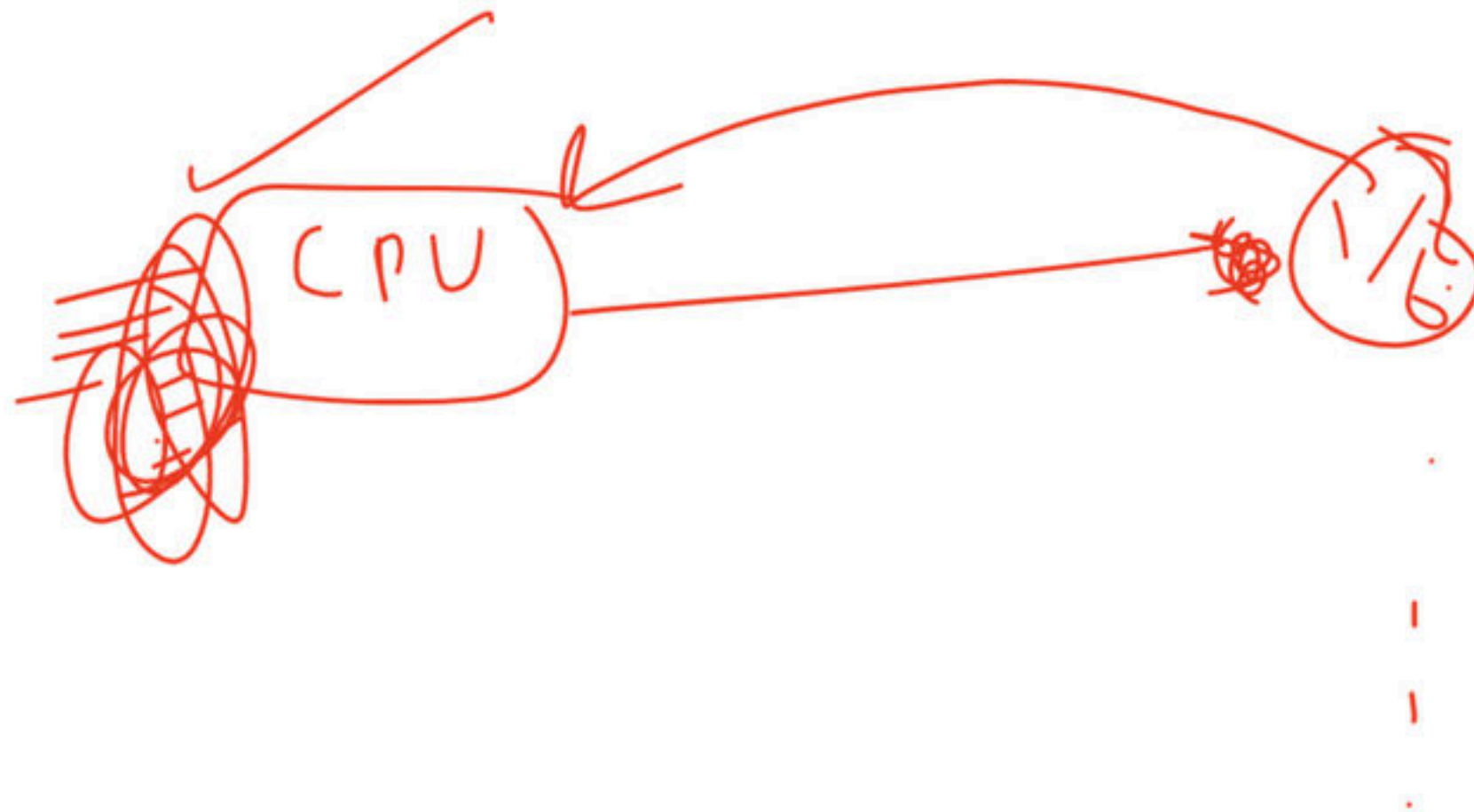


- The best known example of a PC device that uses programmed I/O is the ATA interface
- Parallel ATA (PATA), originally AT Attachment, is an interface standard for the connection of storage devices such as hard disk drives, floppy disk drives, and optical disc drives in computers.



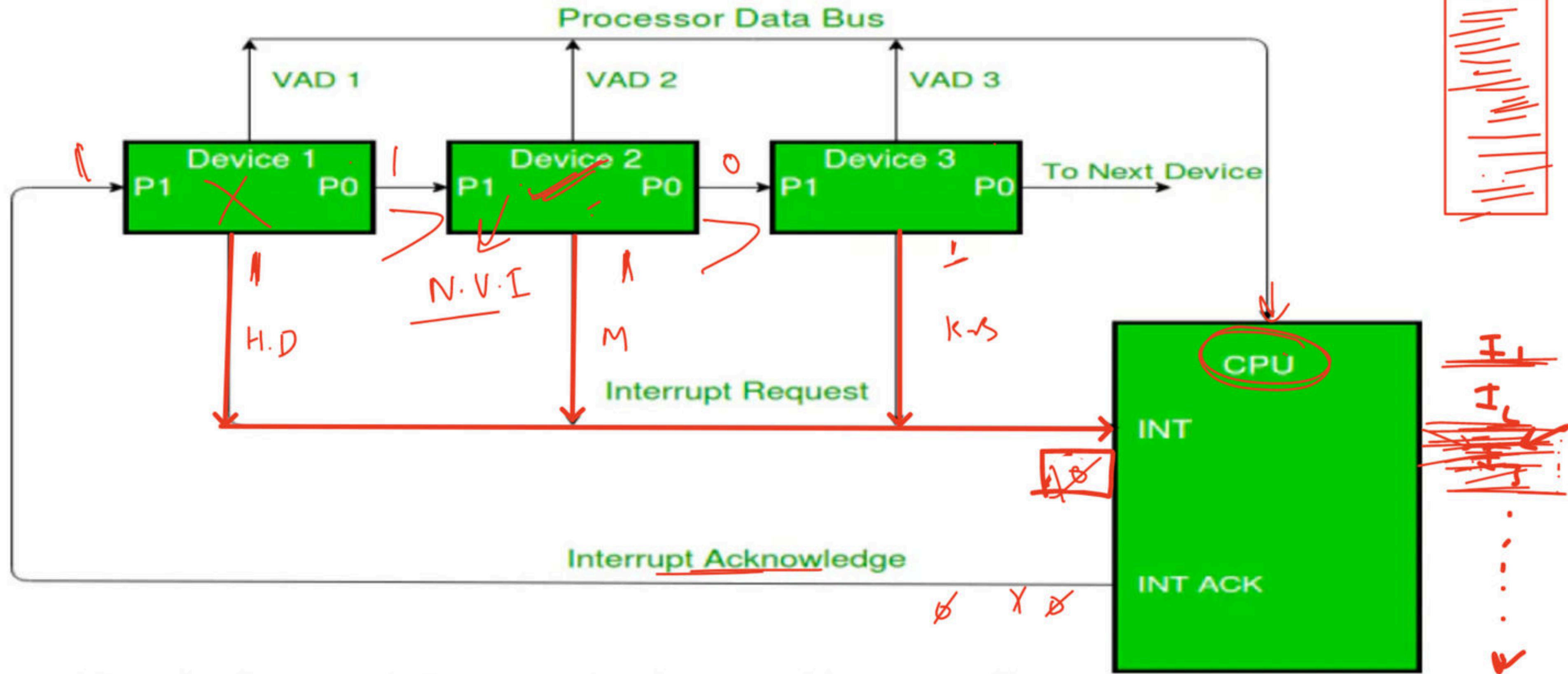
Interrupt initiated i/o

- In this method I/o device interrupt CPU when it is ready for data transfer.
- CPU keep executing instructions and after executing one instruction and before starting another instructions CPU wait and see if there is interrupt or not and if there is interrupt then takes a decision whether CPU should entertain this interrupt or continue with the execution.
- **Note:** - instruction is always absolute in nature and there is nothing like partial execution of instruction.
- The method of handling interrupts of different i/o devices are different, therefor every device has a program or routine called ISR (interrupt service routine) which tell CPU how the interrupt should be managed (and it saves CPU time).



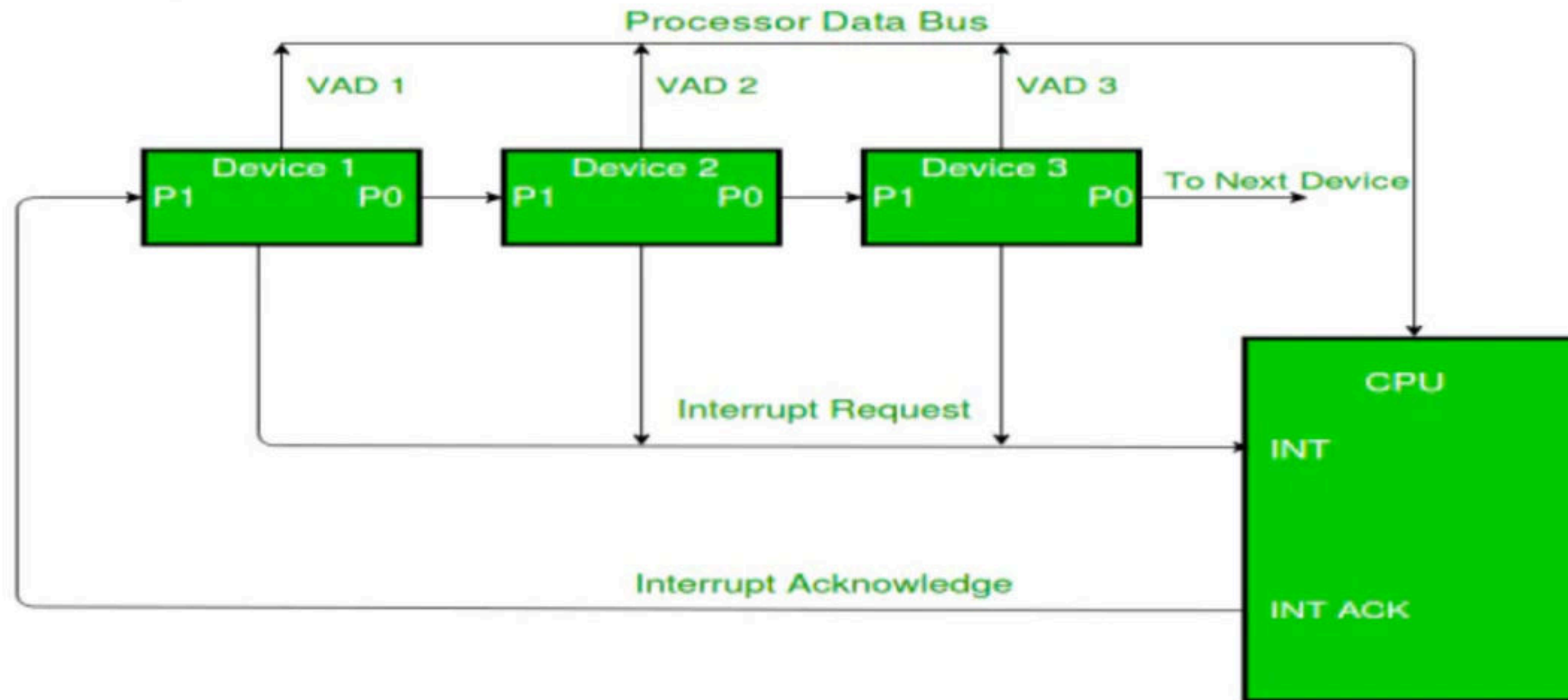
- Interrupt can be of two types: -
 - **Non-vector interrupt:** - Here there is a mutual understanding between CPU and device that where this routine is stored in memory (high priority device)
 - **Vector interrupt:** - Some interrupts may be vectored where the interrupting device will also tell the address that where this routine is stored in the memory.
- It is possible that different i/o devices interrupt at the same time. Now CPU must have a priority decision that which interrupt should be service first and which should be service later.

- H/w solution: - It can be serial or parallel, serial solution is known as Daisy Chaining is a h/w solution which is used to decide priority among different i/o devices (VAD- vector address device)

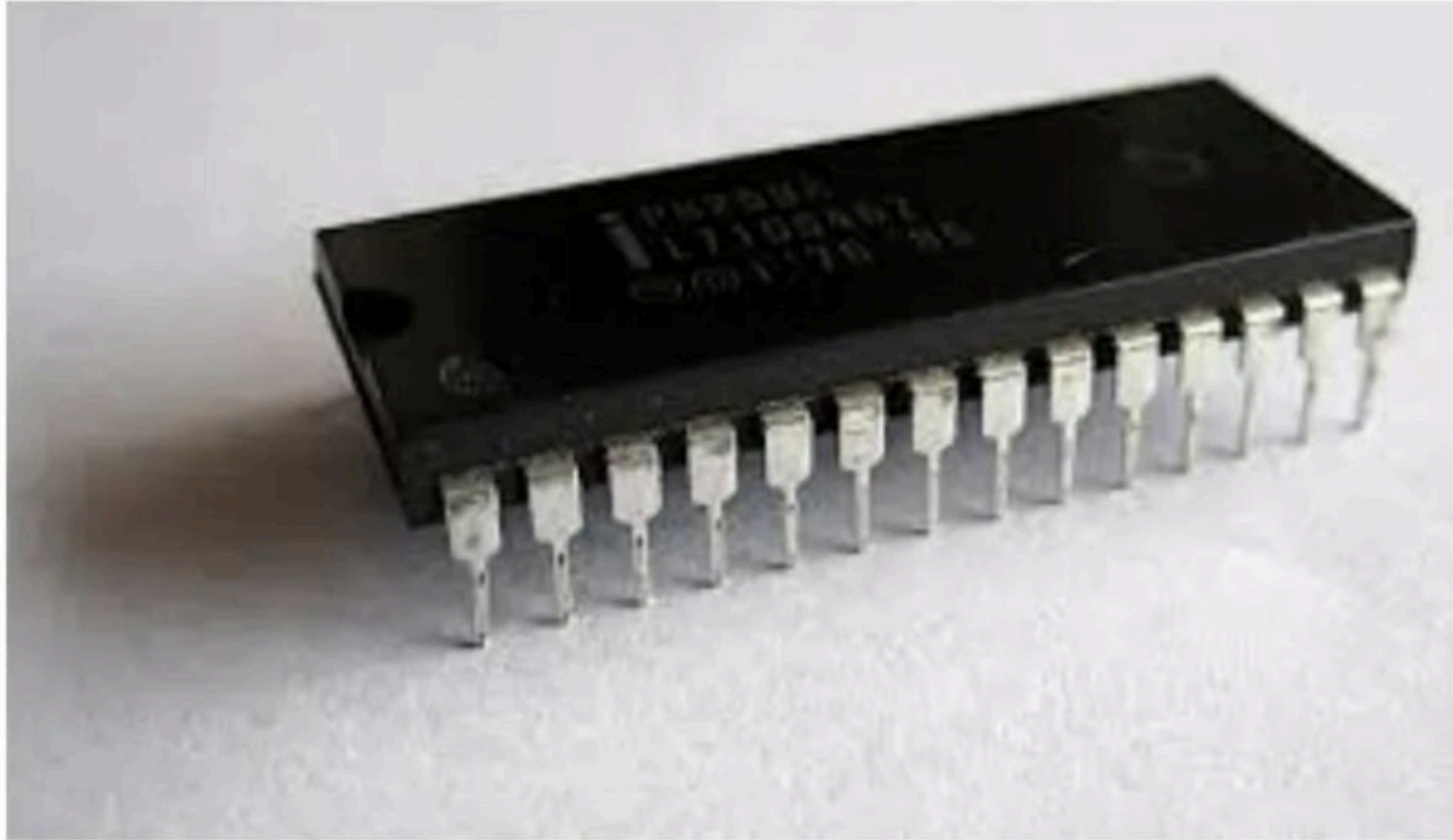


- Out of all possible devices 1 or more device may send an interrupt with a common line.
- When CPU completes 1 instruction and check interrupt in line and found an interrupt, then CPU will enable interrupt acknowledgment line to 1.

- The i/o device placed first in the arrangement will always get acknowledgement first. If it wants to perform i/o then it will put 0 on priority output and will put the address of its interrupt service routine (ISR) on the vector address line.
- If device do not want to perform i/o then will set priority output as 1 and will give chance to the second device, and the processor continue.
- **Advantage:** - very simple, easy to use, easy to understand, relatively fast.
- **Disadvantage:** - here priority fixed and even in case of requirement we cannot change it.



- Intel 82C59A



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Q The following are some events that occur after a device controller issues an interrupt while process L is under execution. **(GATE-2018) (2 Marks)**

(P) The processor pushes the process status of L onto the control stack.

~~**(Q)** The processor finishes the execution of the current instruction.~~

(R) The processor executes the interrupt service routine.

(S) The processor pops the process status of L from the control stack.

(T) The processor loads the new PC value based on the interrupt.

(A) QPTRS

~~**(B)** PTRSQ~~

~~**(C)** TRPQS~~

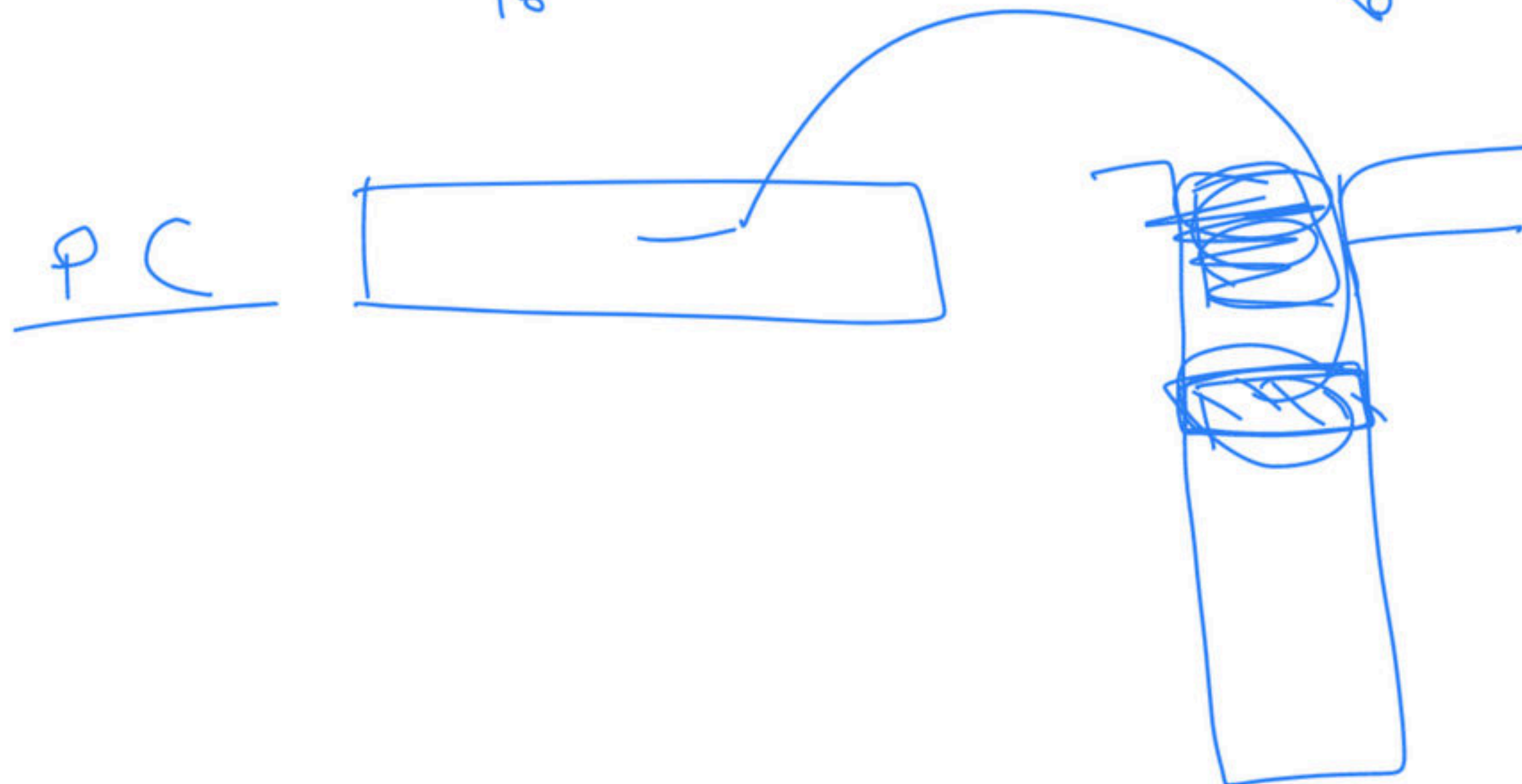
(D) QTPRS

✓
59

16

8

23



Q A CPU handles interrupt by executing interrupt service subroutine _____. (NET-DEC-2015)

- (a) ~~By checking interrupt register after execution of each instruction~~ 8
- (b) ~~by checking interrupt register at the end of the fetch cycle~~ 7
- (c) ~~whenever an interrupt is registered~~ 9
- (d) ~~by checking interrupt register at regular time interval~~ 11

Q A CPU generally handles an interrupt by executing an interrupt service routine: **(GATE-2009) (1 Marks)**

- a)** As soon as an interrupt is raised.
- b)** By checking the interrupt register at the end of fetch cycle.
- c)** By checking the interrupt register after finishing the execution of the current instruction.
- d)** By checking the interrupt register at fixed time intervals.

Q For the daisy chain scheme of connecting I/O devices, which of the following statements is true? **(GATE-1996) (1 Marks)**

- a)** It gives non-uniform priority to various devices
- b)** It gives uniform priority to all devices
- c)** It is only useful for connecting slow devices to a processor device
- d)** It requires a separate interrupt pin on the processor for each device

Q A device with data transfer rate 10 KB/sec is connected to a CPU. Data is transferred byte-wise. Let the interrupt overhead be 4 microsec. The byte transfer time between the device interface register and CPU or memory is negligible. What is the minimum performance gain of operating the device under interrupt mode over operating it under program-controlled mode? **(GATE-2005) (2 Marks)**

(A) 15

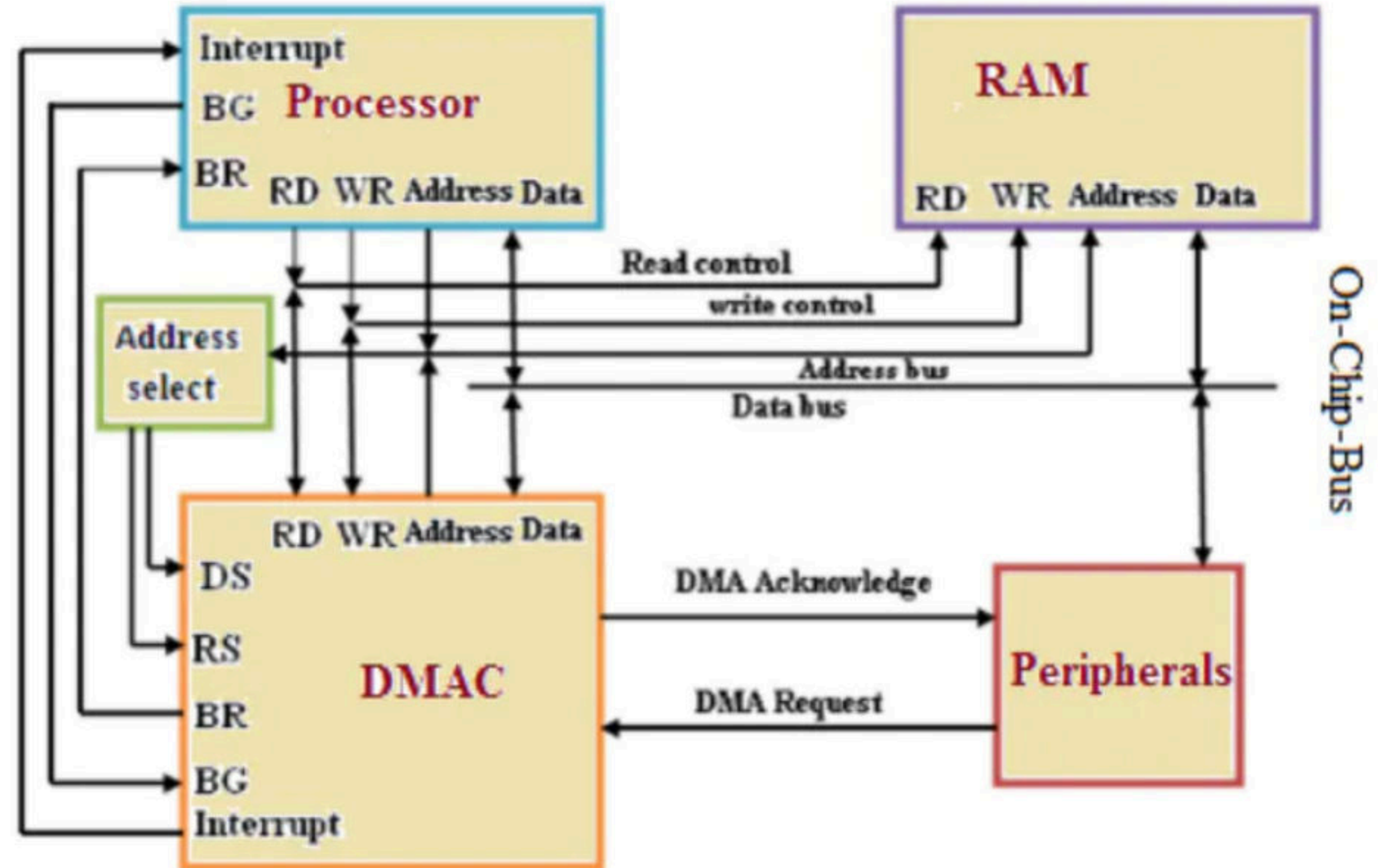
(B) 25

(C) 35

(D) 45

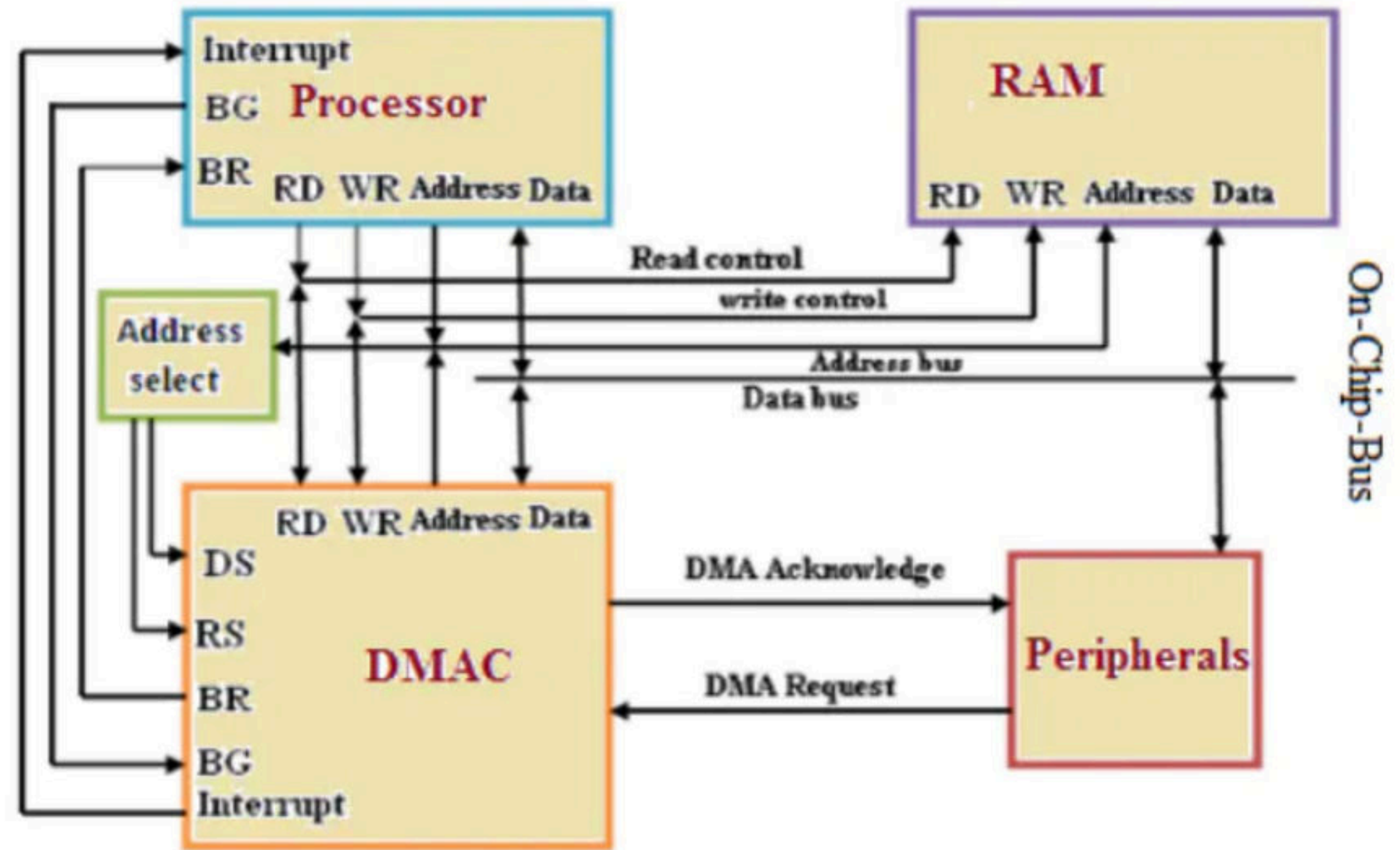
Direct Memory Access (DMA)

- When we want to perform i/o operations then the actual source or destination is either i/o device or memory, but CPU is placed in between just to manage and control the transfer.
- DMA** is the idea where we use a new device is call DMA controller using which CPU allow DMA controller to take control of system buses and perform direct data transfer either from device to memory or from memory to device.



Sequence of DMA transfer: -

- **Step 1:** - I/O device will send a DMA request to DMA controller to perform an i/o operation.
- **Step 2:** - DMA controller will set interrupt and bus request line to 1.
- **Step 3:** - CPU using address bus will select device and registers and then will initiate i/o address register(location), counter (no of words)
- **Step 4:** - CPU will put on the bus grant line to tell DMA controller, now you are the master of system buses.
- **Step 5:** - now DMA controller will put 1 in DMA acknowledgement and using read/write control lines and address line will perform i/o directly between memory and device.



Mode of transfer: -

- **Burst mode:** - when entire i/o transfer is completed and then control comes back to CPU then it is called burst mode transfer. i.e. with high speed device like magnetic disk.
- **Cycle stealing mode:** - When CPU executes an instruction then normally their could following phases.
 - **IF** – Instruction Fetch
 - **ID** – Instruction Decode
 - **OF** – Operand fetch
 - **IX** – Instruction execute
 - **WB** – write back or store result
- Normally in ID and IE phases CPU don't require system buses and if only in that time control is giving to DMA controller then it is called cycle stealing.

- Many hardware systems use DMA, including disk drive controllers, graphics cards, network cards and sound cards.
- In the original IBM PC (and the follow-up PC/XT), there was only one Intel 8237 DMA controller capable of providing four DMA channels (numbered 0–3).

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Q The size of the data count register of a DMA controller is 16 bits. The processor needs to transfer a file of 29,154 kilobytes from disk to main memory. The memory is byte addressable. The minimum number of times the DMA controller needs to get the control of the system bus from the processor to transfer the file from the disk to main memory is _____ **(GATE-2016) (2 Marks)**

Q A hard disk with a transfer rate of 10 Mbytes/ second is constantly transferring data to memory using DMA. The processor runs at 600 MHz, and takes 300 and 900 clock cycles to initiate and complete DMA transfer respectively. If the size of the transfer is 20 Kbytes, what is the percentage of processor time consumed for the transfer operation? **(GATE-2004) (2 Marks)**

(A) 5.0%

(B) 1.0%

(C) 0.5%

(D) 0.1%

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Q A DMA controller transfers 32-bit words to memory using cycle stealing. The words are assembled from a device that transmits characters at a rate of 4800 characters per second. The CPU is fetching and executing instructions at an average rate of one million instructions per second. By how much will the CPU be slowed down because of the DMA transfer? **(NET-DEC-2015)**

(a) 0.6% **(b)** 0.12% **(c)** 1.2% **(d)** 2.5%

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Q Consider a 32-bit microprocessor, with a 16-bit external data bus, driven by an 8 MHz input clock. Assume that this microprocessor has a bus cycle whose minimum duration equals four input clock cycles. What is the maximum data transfer rate for this microprocessor? (**NET-JUNE-2015**)

- a) 8×10^6 bytes/sec
- c) 16×10^6 bytes/sec

- b) 4×10^6 bytes/sec
- d) 4×10^9 bytes/sec

Q On a non-pipelined sequential processor, a program segment, which is a part of the interrupt service routine, is given to transfer 500 bytes from an I/O device to memory.

Initialize the address register

Initialize the count to 500

LOOP: Load a byte from device

Store in memory at address given by address register

Increment the address register

Decrement the count

If count $\neq 0$ go to LOOP

Assume that each statement in this program is equivalent to machine instruction which takes one clock cycle to execute if it is a non-load/store instruction. The load-store instructions take two clock cycles to execute. The designer of the system also has an alternate approach of using DMA controller to implement the same transfer. The DMA controller requires 20 clock cycles for initialization and other overheads. Each DMA transfer cycle takes two clock cycles to transfer one byte of data from the device to the memory. What is the approximate speedup when the DMA controller-based design is used in place of the interrupt driven program-based input-output?

(GATE-2011) (2 Marks)

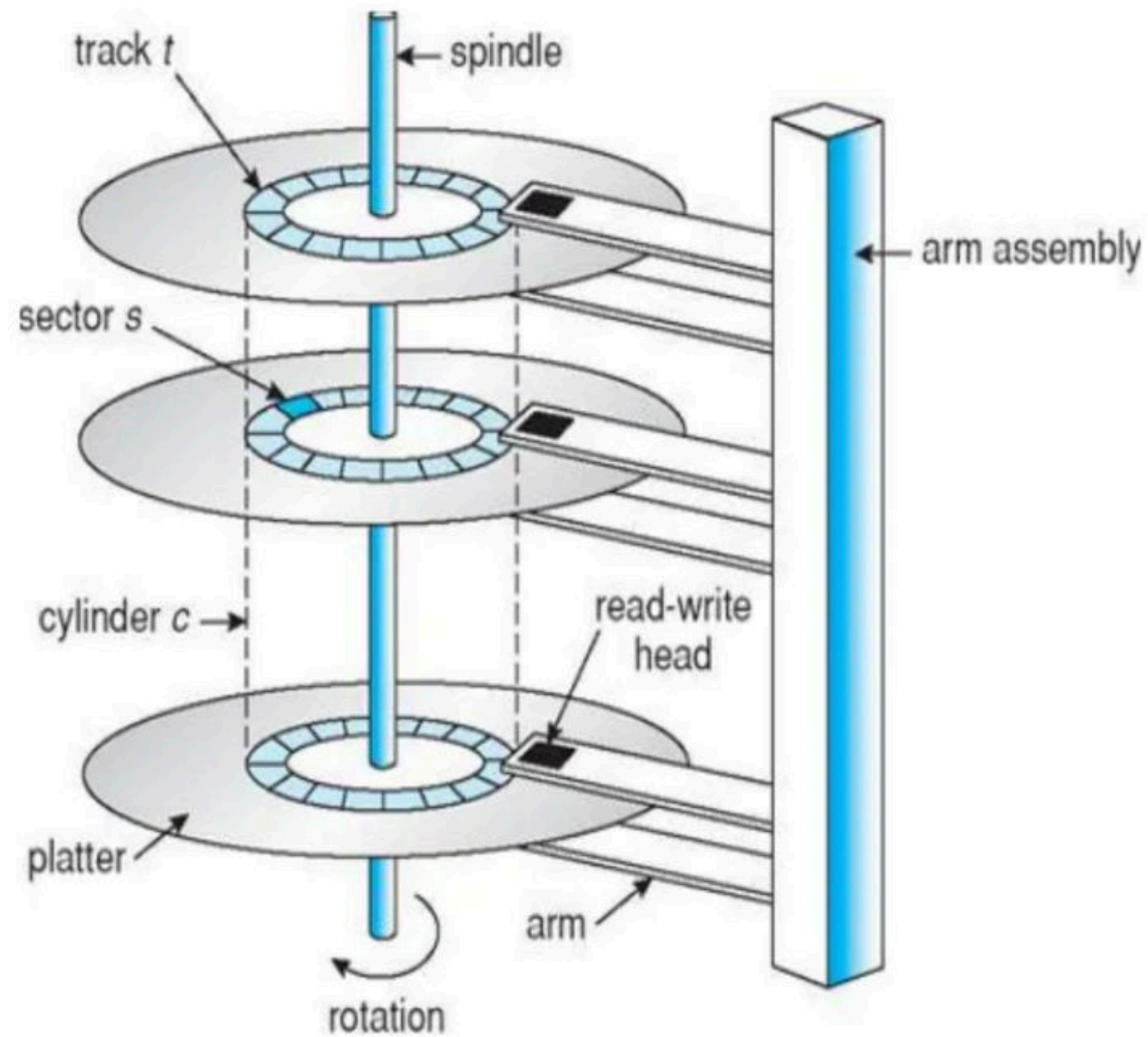
(A) 3.4

(B) 4.4

(C) 5.1

(D) 6.7

Q.1 Consider a computer system with DMA support. The DMA module is transferring one 8-bit character in one CPU cycle from a device to memory through cycle stealing at regular intervals. Consider a 2 MHz processor. If 0.5% processor cycles are used for DMA, the data transfer rate of the device is _____ bits per second. **(GATE-2021)**



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- Total Transfer Time = Seek Time + Rotational Latency + Transfer Time
- **Seek Time**: - It is a time taken by Read/Write header to reach the correct track. (Always given in question)
- **Rotational Latency**: - It is the time taken by read/Write header during the wait for the correct sector. In general, it's a random value, so far average analysis, we consider the time taken by disk to complete half rotation.
- **Transfer Time**: - it is the time taken by read/write header either to read or write on a disk. In general, we assume that in 1 complete rotation, header can read/write the either track, so
- total time will be = $(\text{File Size} / \text{Track Size}) \times \text{time taken to complete one revolution}$.

Q Consider a disk pack with 16 surfaces, 128 tracks per surface and 256 sectors per track. 512 bytes of data are stored in a bit serial manner in a sector. The capacity of the disk pack and the number of bits required to specify a particular sector in the disk are respectively: **(GATE-2007) (1 Marks)**

(A) 256 Mbyte, 19 bits

(C) 512 Mbyte, 20 bits

(B) 256 Mbyte, 28 bits

(D) 64 Gbyte, 28 bit

Q For a magnetic disk with concentric circular tracks, the seek latency is not linearly proportional to the seek distance due to **(GATE-2008) (2 Marks)**

- (A)** non-uniform distribution of requests
- (B)** arm starting and stopping inertia
- (C)** higher capacity of tracks on the periphery of the platter
- (D)** use of unfair arm scheduling policies

Q If the disk is rotating at 360 rpm, determine the effective data transfer rate which is defined as the number of bytes transferred per second between disk and memory. (track size = 512bytes) **(GATE-1995) (2 Marks)**

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Q Consider a typical disk that rotates at 15000 rotations per minute (RPM) and has a transfer rate of 50×10^6 bytes/sec. If the average seek time of the disk is twice the average rotational delay and the controller's transfer time is 10 times the disk transfer time, the average time (in milliseconds) to read or write a 512 byte sector of the disk is _____ **(GATE-2015) (2 Marks)**

Q Consider a disk pack with a seek time of 4 milliseconds and rotational speed of 10000 rotations per minute (RPM). It has 600 sectors per track and each sector can store 512 bytes of data. Consider a file stored in the disk. The file contains 2000 sectors. Assume that every sector access necessitates a seek, and the average rotational latency for accessing each sector is half of the time for one complete rotation. The total time (in milliseconds) needed to read the entire file is _____. **(GATE-2015) (1 Marks)**

Q A certain moving arm disk storage, with one head, has the following specifications:

Number of tracks/recording surface = 200

Disk rotation speed = 2400 rpm

Track storage capacity = 62,500 bits

The average latency of this device is P ms and the data transfer rate is Q bits/sec. Write the values of P and Q .

(GATE-1993) (2 Marks)

Q A hard disk system has the following parameters: **(GATE-2007) (2 Marks)**

Number of tracks = 500

Number of sectors / track = 100

Number of bytes / sector = 500

Time taken by the head to move from one track to adjacent track = 1 ms, Rotation speed = 600 rpm. What is the average time taken for transferring 250 bytes from the disk?

(A) 300.5 ms

(B) 255.5 ms

(C) 255.0 ms

(D) 300.0 ms

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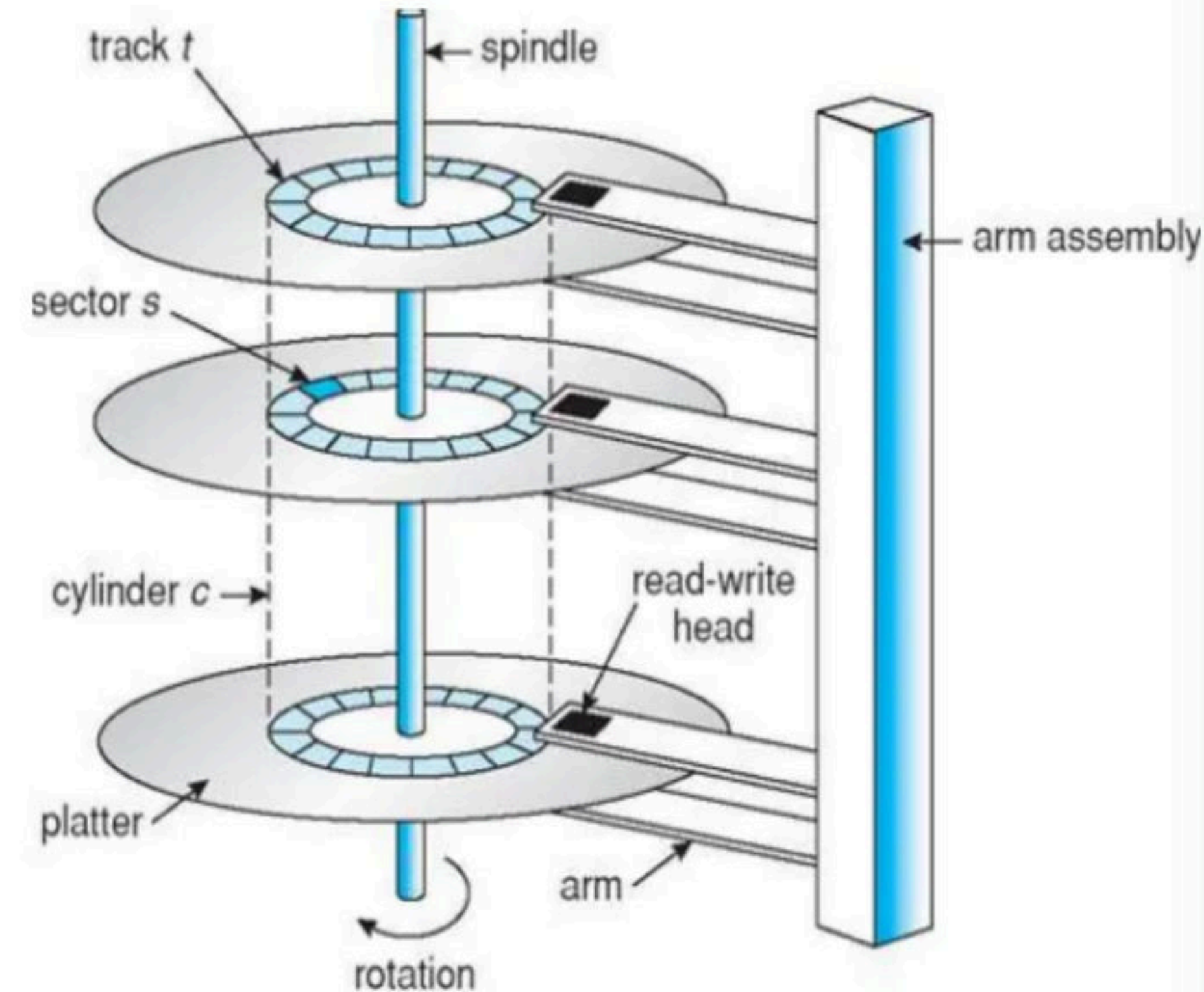
Q A hard disk has 63 sectors per track, 10 platters each with 2 recording surfaces and 1000 cylinders. The address of a sector is given as a triple (c, h, s) , where c is the cylinder number, h is the surface number and s is the sector number. Thus, the 0th sector is addressed as $(0, 0, 0)$, the 1st sector as $(0, 0, 1)$, and so on. The address $\langle 400, 16, 29 \rangle$ corresponds to sector number: **(GATE-2009) (2 Marks)**

(A) 505035

(B) 505036

(C) 505037

(D) 505038



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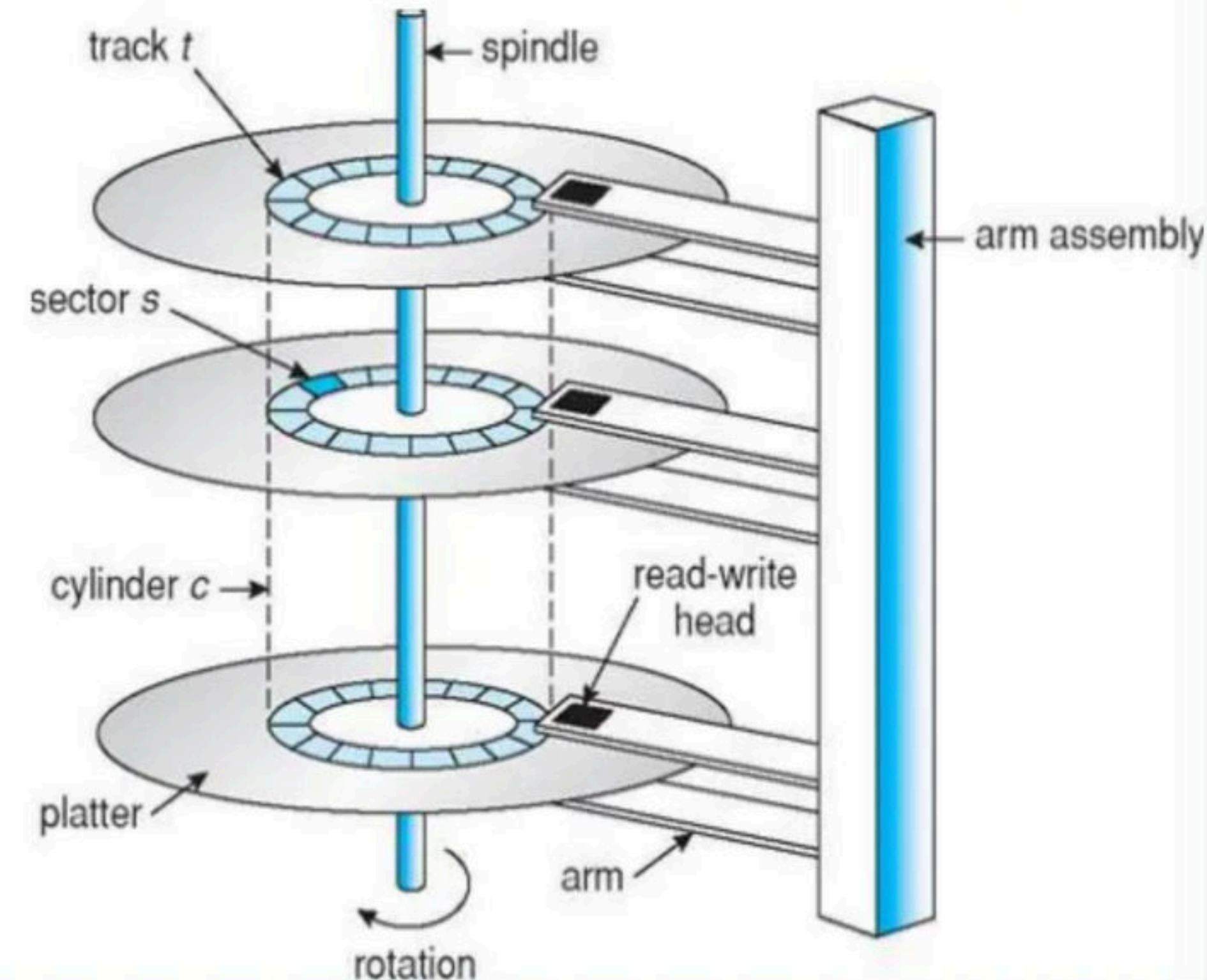
Q Consider a hard disk with 16 recording surfaces (0-15) having 16384 cylinders (0-16383) and each cylinder contains 64 sectors (0-63). Data storage capacity in each sector is 512 bytes. Data are organized cylinder-wise and the addressing format is <cylinder no., surface no., sector no.>. A file of size 42797 KB is stored in the disk and the starting disk location of the file is <1200, 9, 40>. What is the cylinder number of the last sector of the file, if it is stored in a contiguous manner? **(GATE-2013) (1 Marks)**

(A) 1281

(B) 1282

(C) 1283

(D) 1284



Q Consider a disk with 16384 bytes per track having a rotation time of 16 msec and average seek time of 40 msec. What is the time in msec to read a block of 1024 bytes from this disk? **(NET-DEC-2015)**

a) 57 sec

b) 49 sec

c) 48 sec

d) 17 sec

Q An application loads 100 libraries at start-up. Loading each library requires exactly one disk access. The seek time of the disk to a random location is given as 10 milli second. Rotational speed of disk is 6000 rpm. If all 100 libraries are loaded from random locations on the disk, how long does it take to load all libraries? (The time to transfer data from the disk block once the head has been positioned at the start of the block may be neglected) **(GATE-2011) (2 Marks)**

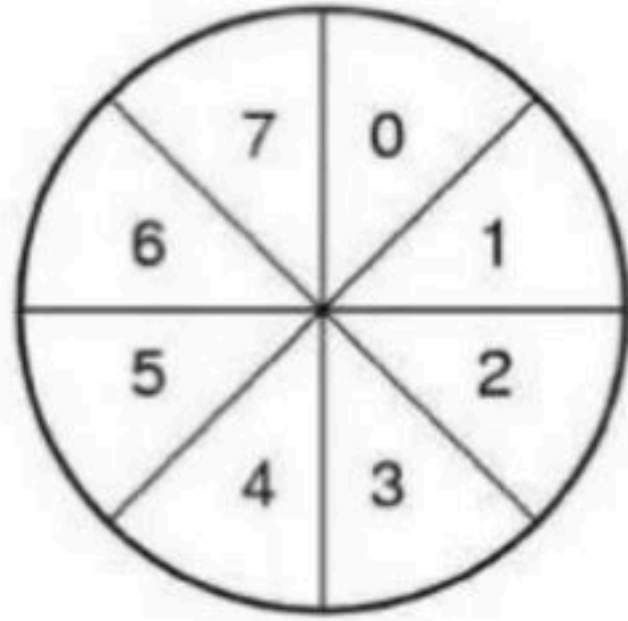
(A) 0.50 s

(B) 1.50 s

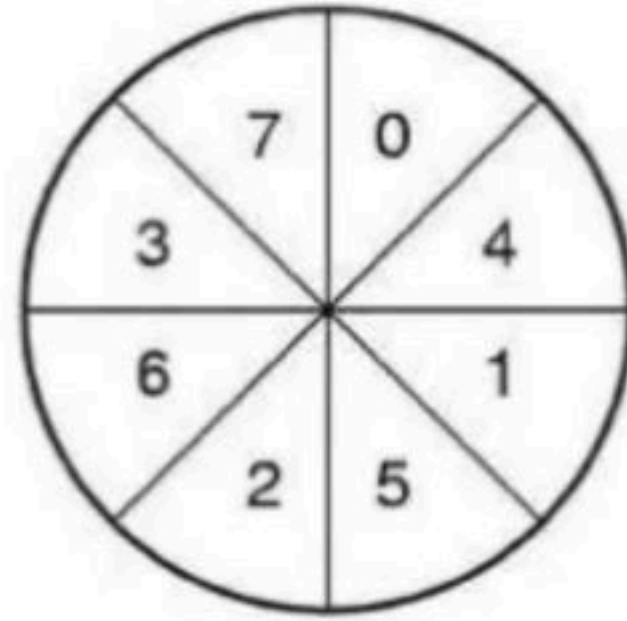
(C) 1.25 s

(D) 1.00 s

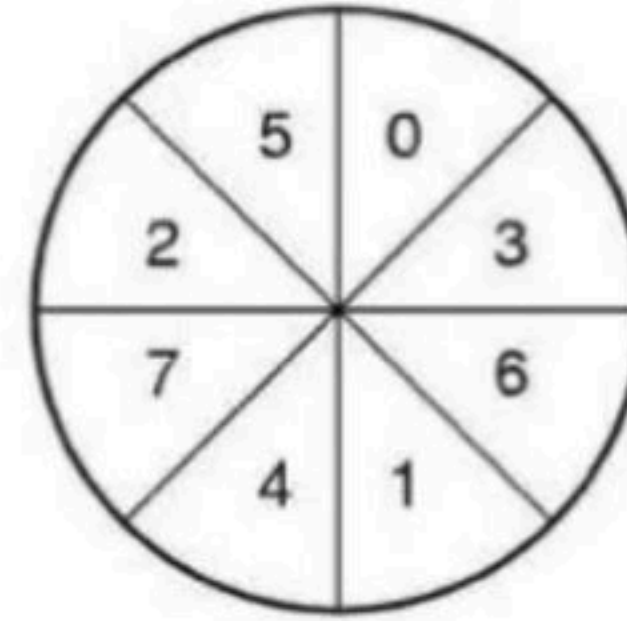
- **Single interleaving:** - in 2 rotation we read 1 track
- **Double interleaving:** - in 2.75 rotation we read 1 track



(a)



(b)



(c)

- No interleaving
- Single interleaving
- Double interleaving